# A zero-dimensional calibration of field redefinitions for AQFT and pAQFT: the Alpöge–Mathew counterexample and uniqueness of its defect

Christoph Mayer*

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**Abstract**

The Alpöge–Mathew map $F : \mathbb{C}^3 \to \mathbb{C}^3$ (July 2026) has constant Jacobian determinant $-2$ but is not injective, disproving the Jacobian Conjecture for $n \geq 3$. Read as a zero-dimensional QFT — $F(\varphi) = J$ a classical field equation, the Bass–Connell–Wright tree expansion its perturbation theory — it is an exactly solvable toy model in which loops are trivial, the tree series *converges*, and the theory nevertheless fails globally: two of three solution sheets sit at infinity over the perturbative vacuum and are invisible to every order of perturbation theory. Our primary interest is what this calibrates for rigorous QFT, especially algebraic QFT (AQFT) and perturbative algebraic QFT (pAQFT). Exact results, verified in a public repository: (i) the pullback $F^*$ is a monomorphism but not an automorphism of the polynomial observable algebra — $\{1, x, x^2\}$ is a basis over $\mathrm{Frac}(\mathrm{im}\, F^*)$, and $x$ is not integral over $\mathrm{im}\, F^*$ (escape-curve certificate $\Leftrightarrow$ non-properness), the cleanest 0D vindication of the Haag–Kastler preference for algebras over field coordinates; (ii) this calibrates the formal deferral in pAQFT (Fredenhagen–Rejzner): "upgrade formal to convergent" is *not* the missing step to a non-perturbative construction — properness of the classical dynamics is; (iii) an exact reduced Keller identity for the weight family $(1, -1, -m)$; (iv)–(v) complete $v$-linear classifications in the sibling systems $(1, -1, -3)$ and $(2, -1, -3)$, culminating in a uniqueness theorem: for every $m \geq 3$ the $v$-linear class of $(1, -1, -m)$ contains only tame automorphisms, so the defect is rigidly attached to the $m = 2$ numerology; (vi) the 0D Witten index (an exact Mathai–Quillen localization) equals $-N(J)$ and jumps $-1 \leftrightarrow -3$ across the wall — index non-invariance is non-properness; (vii) the theory admits no action in three fields in the strongest affine sense, yet its first-order action $W_6 = \bar{\varphi} \cdot F(\varphi)$ is an *explicit* symmetric (variational) Jacobian counterexample in dimension 6 — to our knowledge the first written down — and is necessarily non-coercive; (viii) the global data are packaged as a finite dictionary of *classical-map invariants*, which tensor exactly under ultralocal products, and a certified homotopy-continuation computation shows that the real chamber function of the two-site lattice deformation remains non-constant at every probed kinetic coupling — wall-crossing survives kinetic mixing, with escape real but pointwise on the probed segment and all chamber walls fold-type (exact) — and in the finite-mode $D = 1$ Mathai–Quillen model the index jump survives the path measure exactly (the mode determinant is a positive spectator); (ix) the 0D Buchholz–Fredenhagen caricature is carried out (split verdict): causal factorization is provably vacuous, but the dynamical relation forces the bundle of fiber algebras in which wall, sheet count, and $S_3$ holonomy have exact algebraic addresses. We state honestly what the model does *not* imply (nothing about UV renormalization, Borel summability, or $D = 4$ existence). AI-assisted exploratory research, not peer reviewed; every claim is labelled exact, box-checked, or open, and is reproducible from a cited script.

*Independent researcher. Email: hcab14@gmail.com.

# Contents

# 1 Introduction

## 1.1 Background

The Jacobian Conjecture (Keller, 1939 [2]) asserted that every polynomial map $F : \mathbb{C}^n \to \mathbb{C}^n$ with constant nonzero Jacobian determinant (a *Keller map*) is a polynomial automorphism. Despite the degree-reduction theory of Bass–Connell–Wright [3], the formal tree-inversion formulas of Wright [4], and decades of partial results collected in van den Essen's monograph [5], the conjecture remained open in every dimension $n \geq 2$ until July 19, 2026, when Levent Alpöge announced (with Akhil Mathew; the map itself was produced by a large language model in response to Mathew's question) an explicit counterexample in dimension 3 [1]: the map

$$F(x, y, z) = \big((1+xy)^3 z + y^2(1+xy)(4+3xy),\ \ y+3x(1+xy)^2 z+3xy^2(4+3xy),\ \ 2x-3x^2y-x^3z\big) \tag{1}$$

satisfies $\det DF \equiv -2$ yet identifies the three points $(0, 0, -\frac{1}{4})$, $(1, -\frac{3}{2}, \frac{13}{2})$, $(-1, \frac{3}{2}, \frac{13}{2})$, all mapping to $(-\frac{1}{4}, 0, 0)$. Hence the conjecture is false for all $n \geq 3$ (adjoin identity coordinates); the case $n = 2$ remains open. The example was verified by the community and formalized in Lean 4; all claims about it in this paper are independently re-verified in the repository accompanying this paper (`scripts/verify_counterexample.py`).

The Jacobian Conjecture has a well-known reformulation in zero-dimensional perturbative quantum field theory, made precise by Abdesselam [6] on the basis of the tree formulas of [3, 4] and the constructive combinatorics of Abdesselam–Rivasseau [7]: $F(\varphi) = J$ is the classical field equation of a three-component scalar model in $D = 0$ spacetime dimensions with external source $J$; the linearization $DF(0)$ is the inverse propagator; the nonlinear monomials of $F$ are interaction vertices; and the perturbative inversion $\varphi(J)$ is exactly the sum over rooted tree Feynman graphs. Constant Jacobian determinant means all loop corrections are field-independent constants: the theory looks free at loop level. A genuine counterexample therefore provides, for the first time, an exactly solvable toy model in which perturbation theory is well defined at every order, loop corrections are trivial, and the theory still fails globally.

*Why this matters for AQFT and pAQFT.* The motivation of the present paper is not primarily to enlarge the catalogue of Jacobian counterexamples, but to use this toy model as a *calibration device* for rigorous QFT — especially the algebraic tradition of Haag–Kastler [8] and the perturbative algebraic QFT (pAQFT) framework of Fredenhagen–Rejzner [9, 10, 11]. Polynomial field redefinitions with $\det DF = \text{const} \neq 0$ are used throughout renormalization theory; invertibility is usually established at the formal-power-series level, where it is automatic. The counterexample shows that, already with three field components, a polynomial redefinition with constant Jacobian need not be globally injective: the formal inverse exists, *converges* near the vacuum, and is still only a local section of a 3-sheeted covering. The exact structure of the observable algebra $\operatorname{im} F^*$ (Section 3) turns the Haag–Kastler preference for algebras over field coordinates into a theorem-sized 0D fact; the pAQFT deferral of convergence and globality is calibrated rather than contradicted (Section 11). The sibling searches and the uniqueness theorem then ask how rigid this defect is: is it a generic feature of constant-Jacobian dynamics, or a delicate coincidence of the $m = 2$ weight arithmetic?

## 1.2 Contents and main results

This paper condenses, into a self-contained account, the exact-structure results obtained in the public repository

https://github.com/hcab14/JacobianConjectureQFT

about the Alpöge–Mathew map and its two nearest $\mathbb{C}^*$-equivariance siblings, with the QFT/AQFT/pAQFT reading as the primary interpretive frame. The main results are:

- **Section 2** sets up the $D = 0$ QFT dictionary for the map (1): the first-order (conjugate-field) action, the $\mathbb{C}^*$-grading with source weights $(1, -1, -2)$, the escape locus $\{p = 0\}$ (the non-properness set), and the reduction of $\det DF$ to the two-dimensional invariant plane.
- **Section 3** gives the exact structure of the observable algebra $\operatorname{im} F^*$: $\mathbb{C}(x, y, z)$ is free with basis $\{1, x, x^2\}$ over the fraction field of $\operatorname{im} F^*$, every polynomial observable has a unique computable normal form $c_0(F) + c_1(F)x + c_2(F)x^2$, and $x$ is *not integral* over $\operatorname{im} F^*$ — an exact escape-curve certificate showing that no finite-module refinement is possible (Theorems 3.1, 3.2).
- **Section 4** states the exact reduced Keller identity for the whole weight family $(1, -1, -m)$: $\det DF = \det M$ and $R^m \det M = J_2(PR^m, QR)$, reducing the 3D Keller problem to two variables (Proposition 4.1).
- **Section 5** treats the weight system $(2, -1, -3)$, whose invariant ring is an $A_1$ quadric cone: a chart-based reduced Keller identity, a complete classification of the degree-1 Keller box (two families of tame automorphisms, Theorem 5.1), provable emptiness of the $\mathbb{Z}_2$ and $\mathbb{Z}_3$ stabilizer-jump mechanisms in two boxes, and an honest account of the computational wall met by the degree-2 box.
- **Section 6** classifies the $v$-linear class of $(1, -1, -3)$ for all $w$-degrees by a Wronskian stratification (Theorem 6.2): every Keller map is a tame automorphism; the 3:1 mechanism is empty; the Alpöge–Mathew mechanism is obstructed at $m = 3$ by an exact numerological incompatibility (Section 6.6). One substratum ($D_3$ with non-squarefree data) is a Gröbner-box gap (Section 6.4).
- **Section 7** is the headline theorem: the $m = 3$ classification extends uniformly to *every* $m \geq 3$ (Theorem 7.1). For $m \geq 4$ the D-strata are empty with no box at all;

Alpöge–Mathew survives at $m = 2$ exactly where the killing factor $u^{m-2}$ trivializes (`scripts/search_11m.py`, `docs/SEARCH_11M.md`).

- **Section 8** adds the supersymmetric and variational faces of the defect. The 0D Witten index (signed solution count, realized by an exact Mathai–Quillen localization) equals $-N(J)$ and *jumps* $-1 \leftrightarrow -3$ across the wall — index non-invariance is non-properness made quantitative. And the variational question is settled constructively: the Alpöge–Mathew defect is non-variational in dimension 3 in the strongest affine sense, yet its cotangent lift $W_6 = \bar{\varphi} \cdot F(\varphi)$ — the first-order action itself — is an *explicit* symmetric Jacobian counterexample in dimension 6 ($\det \operatorname{Hess} W_6 \equiv -4$, rational witnesses, Theorem 8.2), necessarily non-coercive.

- **Section 9** packages the results as a finite dictionary of *classical-map invariants* (I1–I8: fiber degree, monodromy, Jelonek divisor, chamber function, signed index, observable defect, torus weights, variationality) and takes the first step off $D = 0$: the invariants tensor exactly under ultralocal products, linear kinetic mixing provably does not restore properness, and a certified homotopy-continuation computation shows the real chamber function of the two-site lattice deformation stays *non-constant* at every probed coupling — wall-crossing survives kinetic mixing, with real solutions arriving from infinity at finite coupling. Exact walls on the probed segment (Section 9.4): all chamber walls are fold-type, escape is pointwise; the HC "odd jumps" were completeness artifacts. In the finite-mode $D = 1$ Mathai–Quillen model the index jump survives the path measure: the mode fluctuation determinant at every equilibrium is an exactly positive, $J$-independent spectator (Section 9.5).

- **Section 10** carries out the 0D Buchholz–Fredenhagen caricature (split verdict): causal factorization is provably vacuous on a point, but the dynamical relation survives and forces the bundle of fiber algebras $A_J = \mathbb{C}[x, y, z]/(F - J)$, in which the wall is the rank-jump locus and pole divisor, the $1 \leftrightarrow 3$ count is the character count, and $S_3$ is the transport holonomy (with provably *no* deck action); any sector-resolving $S(J)$ is multi-valued or singular on the wall.

- **Section 11** is the interpretive core: what the model calibrates for AQFT (Haag–Kastler), pAQFT (Fredenhagen–Rejzner), and non-perturbative algebraic constructions (Buchholz–Fredenhagen), together with an explicit ledger of what it does *not* imply; the uniqueness theorem is read as rigidity of the non-properness defect.

- **Sections 12 and 13** give QFT-oriented next probes and the reproducibility table.

### 1.3 Provenance and verification

This paper reports **AI-assisted exploratory research**: it was produced in interactive sessions with AI coding agents (large language model agents wrote and ran the verification scripts), and it has **not been peer reviewed**. The counterexample itself is due to Alpöge and Mathew [1]; this work only studies it. The claims should be read with the following calibration, which the text states case by case:

- *Exact / symbolic results* (the bulk of the paper): every displayed identity and every emptiness claim stated as proved is verified by exact symbolic computation — sympy [22] over

$\mathbb{Q}$, and Gröbner-basis certificates computed with msolve [19] and Singular [21] — in the cited scripts of the public repository. "Verified" means the script passes, within the stated scope; the scope lines (often box- or chart-limited) matter and are always given.

- *Numerical results* (e.g. the $S_3$ monodromy of the three sheets, rigidity continuation): labelled as evidence, not proof.
- *Interpretation* (the QFT and anomaly readings): clearly flagged as such.

Every theorem and proposition below cites the repository script that asserts it (for the main theorem this is `scripts/search_113.py`); Section 13 lists the commands and runtimes needed to reproduce all of them. Corrections and counter-arguments are welcome.

### 1.4 Notation

Throughout, $F = (F_1, F_2, F_3) : \mathbb{C}^3 \to \mathbb{C}^3$ is a polynomial map in source coordinates $\varphi = (x, y, z)$ with target (source-of-the-QFT) coordinates $J = (a, b, c)$. $DF$ is the Jacobian matrix and a *Keller map* has $\det DF \equiv \kappa \in \mathbb{C}^\times$. For a weight system $(1, -1, -m)$ we write

$$w = xy, \qquad v = x^m z$$

for the generators of the invariant ring, and

$$J_2(A, B) \;=\; \partial_w A\, \partial_v B - \partial_v A\, \partial_w B$$

for the Jacobian of two functions of $(w, v)$. Primes denote $d/dw$. For the weight system $(2, -1, -3)$ the invariants are $u_1 = xy^2$, $u_2 = x^2yz$, $u_3 = x^3z^2$ with chart coordinates $(t, p, q)$ as in Section 5. The polynomials

$$p \;=\; 27a^2c^2 - 18abc + b^3c - b^2 + 16a, \qquad D_0 \;=\; 27ac^2 - 9bc + 8$$

are the escape (non-properness) polynomial and the fiber-parametrization discriminant factor of the Alpöge–Mathew map (the symbol $p$ is also used for the chart coordinate $u_1$ in Section 5 and for lowercase coefficient functions $p_0, p_1$ in Section 6; the context always disambiguates).

## 2 The Alpöge–Mathew map as a zero-dimensional QFT

### 2.1 The counterexample and the dictionary

Direct computation (`scripts/verify_counterexample.py`) confirms $\det DF \equiv -2$ for the map (1) and the triple fiber over $(-\frac{1}{4}, 0, 0)$. Note that all three colliding points are real: the restriction $F : \mathbb{R}^3 \to \mathbb{R}^3$ is a real polynomial local diffeomorphism with *constant* Jacobian that fails to be injective.

One would like to write $F(\varphi) = J$ as the stationarity condition of an action $S(\varphi)$, but that is impossible: $F$ is a gradient iff $DF$ is symmetric, and here already $\partial F_1/\partial z|_0 = 1 \neq 2 = \partial F_3/\partial x|_0$ (no affine reframing helps either; see Proposition 8.1). The standard remedy [6] is a conjugate (auxiliary) field $\bar{\varphi} = (\bar{x}, \bar{y}, \bar{z})$ and the first-order action

$$S(\bar{\varphi}, \varphi) = \bar{\varphi} \cdot \big(F(\varphi) - J\big), \qquad Z(J) = \int d^3\bar{\varphi}\, d^3\varphi\; e^{-S(\bar{\varphi}, \varphi)},$$

where the $\bar\varphi$-integral localizes $Z(J)$ on $\delta^3(F(\varphi)-J)$. The ghost sector that would produce $\det DF(\varphi)$ is trivial here, contributing only the constant $-2$. Splitting $S = S_{\text{free}} + S_{\text{int}}$ with $S_{\text{free}} = \bar\varphi\cdot L\varphi - \bar\varphi\cdot J$, $L = DF(0)$, the interaction consists of thirteen vertices, each with exactly one $\bar\varphi$-leg, from cubic ($\bar x\, y^2$, $\bar y\, xz$, $\bar z\, x^2y$) to octic ($\bar x\, x^3y^3z$). The one-point function $\langle\varphi\rangle = G(J)$ is the formal inverse of $F$: Wick contraction generates precisely the Bass–Connell–Wright–Wright sum over rooted trees [3, 4], with no loop corrections beyond the constant $\det DF = -2$.

Two facts about this expansion, established in the repository and used below only as context, are worth recording. First, the tree series *converges*, with finite nonzero radius; it is the Taylor series of one branch of an explicit degree-3 algebraic function: Gröbner elimination (`scripts/branch_locus.py`) shows that the $x$-coordinate of any preimage of $(a,b,c)$ satisfies the eliminant cubic

$$p\,x^3 + q\,x + r = 0, \qquad q = 4-3bc, \qquad r = -2c, \tag{2}$$

with $p$ as in Section 1.4. Second, what fails is *globality*, not summability: on $\{p=0\}$ the cubic drops degree and a solution sheet escapes to infinity in field space. The set $\{p=0\}$ is exactly the non-properness set of $F$ in the sense of Jelonek [15], the perturbative vacuum $J=0$ lies *on* it, and the two non-perturbative sheets over the vacuum sit at infinity, invisible to every order of perturbation theory. Off $\{p=0\}$ the map is a genuine 3-sheeted covering (being étale, sheets never merge at finite points); its geometric monodromy is the full $S_3$ (**Exact**: irreducible eliminant + local Puiseux at a smooth wall point prove wall meridians are transpositions, hence $\mathrm{Mon} = S_3 = \mathrm{Gal}$; representation = canonical $B_3 \twoheadrightarrow S_3$ via the $A_2$ isomorphism of the wall complement — `scripts/certified_monodromy.py`; labelled permutations on a chosen basis remain numerical in `scripts/monodromy.py`).

### 2.2 $\mathbb{C}^*$-equivariance and reduction to the invariant plane

The model is equivariant under the $\mathbb{C}^*$-action $\lambda.(x,y,z) = (\lambda x, \lambda^{-1}y, \lambda^{-2}z)$: the components of $F$ scale with weights $(-2,-1,1)$, so the sources $(a,b,c)$ carry weights $(-2,-1,1)$ and every term of the action has weight 0. The invariant ring of the source action is the *free* polynomial ring $\mathbb{C}[w,v]$ with $w = xy$, $v = x^2z$, and the Alpöge–Mathew map is exactly of the equivariant normal form

$$F = \Big(\frac{P(w,v)}{x^2},\ \frac{Q(w,v)}{x},\ x\,R(w,v)\Big),$$

with

$$P_{\text{AM}} = (1+w)^3v + w^2(1+w)(4+3w), \quad Q_{\text{AM}} = w + 3(1+w)^2v + 3w^2(4+3w), \quad R_{\text{AM}} = 2-3w-v.$$

For such maps $\det DF$ descends to a polynomial in $(w,v)$ alone, so the 3D Keller condition is a 2D problem — this is the special case $m=2$ of Proposition 4.1 below, and it is the engine of the classification results in Sections 5 and 6. The counterexample mechanism is a *stabilizer jump*: the target $a$-axis (weight $-2$) has stabilizer $\mathbb{Z}_2$, so a free source orbit mapping into it (a common zero of $(Q,R)$ with $P\neq 0$) is 2:1 — realized for (1) along the explicit curve $\varphi(s) = (\pm s, \mp\frac{3}{2s}, \frac{13}{2s^2})$ with $F(\varphi(s)) = (-\frac{1}{4s^2}, 0, 0)$.

# 3 The exact structure of the observable algebra $\operatorname{im} F^*$

The pullback $F^*: \mathbb{C}[a,b,c] \to \mathbb{C}[x,y,z]$, $F^*\mathcal{O} = \mathcal{O}\circ F$, is injective ($F$ is dominant) and not surjective ($F$ is 3-to-1): the polynomial observables of the "redefined field" $F(\varphi)$ form a proper subalgebra. This section makes the cokernel completely explicit. All statements are asserted exactly in `scripts/missing_observables.py` ($\sim$15 s).

**Theorem 3.1** (Field-level structure of $\operatorname{im} F^*$; verified)**.** *Let $K = \mathbb{C}(a,b,c) \hookrightarrow L = \mathbb{C}(x,y,z)$ via $F^*$. Then:*

1. *$L = K(x)$: adjoining the single coordinate $x$ gives everything; $y$ and $z$ are explicit $K$-rational expressions in $x$ with denominators involving only $D_0(F)$ (the fiber parametrization of the repository, read backwards).*

2. *$[L : K] = 3$ with basis $\{1, x, x^2\}$: $x$ satisfies the eliminant cubic (2) with coefficients evaluated at $F$, and the cubic is irreducible over $K$.*

3. Normal form: *every polynomial observable $\mathcal{O}(x,y,z)$ is uniquely*

$$\mathcal{O} \;=\; c_0(F) + c_1(F)\,x + c_2(F)\,x^2, \qquad c_i \in \mathbb{C}(a,b,c),$$

   *computable by one polynomial division (reduction mod the eliminant); the denominators of the $c_i$ are powers of $D_0$ times powers of the escape polynomial $p$.*

4. Membership criterion: *$\mathcal{O}$ is a function of the redefined field alone iff $c_1 = c_2 = 0$. The missing observables are, at field level, the free module generated by $\{x, x^2\}$ over* $\operatorname{im} F^*$ *— precisely the two sheet-separating degrees of freedom of the degree-3 covering.*

Worked examples (asserted): $F_1 \mapsto (a, 0, 0)$; $F_3^2 - 5F_2 \mapsto (c^2 - 5b,\, 0,\, 0)$; and

$$y \;=\; \frac{81abc^2 - 72ac - 15b^2c + 16b}{2D_0} \;+\; \frac{3p}{D_0}\,x \;-\; \frac{3(9ac - b)\,p}{2D_0}\,x^2,$$

with a similar expansion for $z$ whose separator coefficients also carry the factor $p$. The escape polynomial appears *verbatim* in the separator coefficients: the amount by which a coordinate fails to be an observable of $F(\varphi)$ is proportional to the non-properness polynomial.

**Theorem 3.2** (No finite-module version; exact escape certificate; verified)**.** *$x$ is not integral over* $\operatorname{im} F^*$*. Along the curve*

$$\varphi(T) \;=\; \Big(T,\; y_0,\; \frac{2T - 3T^2 y_0 - c_3}{T^3}\Big),$$

*the source escapes ($x = T \to \infty$) while $F(\varphi(T))$ converges to the finite target $\big(y_0^2(1 - c_3 y_0),\; y_0(4 - 3c_3 y_0),\; c_3\big)$, which lies exactly on the wall $\{p = 0\}$ (this is a rational parametrization of the Jelonek set by $(y_0, c_3)$). An integral element must stay bounded when its coefficients do, so no monic relation for $x$ over $\mathbb{C}[a,b,c]$ exists; in particular $\mathbb{C}[x,y,z]$ is not a finite module over* $\operatorname{im} F^*$*.*

*Remark* 3.3. This is the ring-theoretic face of non-properness: for these dominant Keller maps, finite module $\Leftrightarrow$ proper map, and a proper Keller map would be injective. Every counterexample to the Jacobian Conjecture must therefore exhibit this infinite-module pathology; the normal form of Theorem 3.1 is the best possible structure statement.

Membership can be decided cheaply: field-level membership by the normal form (one division); strict polynomial membership by the standard tag-variable Gröbner test; and for the coordinates themselves a single exact fiber computation suffices — the fiber over $F(1,2,3)$ consists of three points whose $x$-, $y$- and $z$-coordinates are pairwise distinct (verified), and any element of $\operatorname{im} F^*$ is constant on fibers.

*QFT / AQFT reading.* The observable algebra of the redefined field has corank 2 as a module inside the full observable algebra, with explicit generators $x, x^2$ for the quotient: the operators that distinguish the three classical vacua over a source $J$. They are exactly the observables that perturbation theory around one vacuum constructs as algebraic functions of $J$ (branches of the cubic) and that no polynomial function of $J$ represents; they cease to exist as finite quantities exactly where $p = 0$. "Which observables are missing" and "where does the theory degenerate" have the same answer. In the language of Section 11.2: $F^*$ is a monomorphism but not an automorphism of the observable algebra — the 0D fact that underwrites the Haag–Kastler preference for algebras over field coordinates.

## 4 The reduced Keller identity for the family $(1, -1, -m)$

The reduction used for the Alpöge–Mathew system in Section 2.2 holds uniformly in the weight family $(1, -1, -m)$. It is proved as an identity of differential polynomials (sympy applied to undetermined functions — not a spot check) in `scripts/reduction_113.py`, module `jcqft/reduction_w.py`.

**Proposition 4.1** (Reduced Keller identity; verified for $m \leq 4$, uniform proof)**.** *Fix $m \geq 1$ and the source action $\lambda.(x, y, z) = (\lambda x, \lambda^{-1} y, \lambda^{-m} z)$, with free invariant ring $\mathbb{C}[w, v]$, $w = xy$, $v = x^m z$. Invertibility of $DF(0)$ forces the component weights of an equivariant $F$ to be a permutation of $(-m, -1, 1)$; fixing the order,*

$$F \;=\; \Big(\frac{P(w,v)}{x^m},\; \frac{Q(w,v)}{x},\; x\,R(w,v)\Big).$$

*Then:*

1. *$F$ is a polynomial map iff every monomial $w^j v^k$ of $P$ satisfies $j + mk \geq m$ and every monomial of $Q$ satisfies $j + mk \geq 1$ ($R$ is unconstrained);*

2. $\det DF$ *is a function of $(w, v)$ alone:*

$$\det DF \;=\; \det M \;=\; -m\,P\,J_2(Q,R) + Q\,J_2(P,R) + R\,J_2(P,Q),$$

   *where $M$ is the $3 \times 3$ matrix with rows $(d_i S_i,\ \partial_w S_i,\ \partial_v S_i)$, $(S_1, S_2, S_3) = (P, Q, R)$, $d = (-m, -1, 1)$;*

3. compact form: *$R^m \det M = J_2(PR^m,\, QR)$ identically, so*

$$\det DF \equiv \kappa \quad\iff\quad J_2(PR^m,\, QR) = \kappa\, R^m.$$

*The 3D Keller problem for this equivariance class is therefore a 2D problem, for every $m$.*

*Proof sketch.* In the coordinates $(x, w, v)$ one has $\det \partial(x, w, v)/\partial(x, y, z) = x^{m+1}$, and the equivariant form gives $\det DF = x^{\sum d_i + m+1} \det M = \det M$ since $\sum d_i = -m$. The script proves the identity (and the compact form) with $P, Q, R$ undetermined functions for $m = 1, 2, 3, 4$; the chain-rule computation is uniform in $m$. Consistency at $m = 2$: the machinery reproduces the known reduction on the Alpöge–Mathew map, with residual 0 at $\kappa = -2$. The polynomiality box is verified in both directions on exhaustive monomial boxes at $m = 3$. □

Two gauge structures accompany the reduction and are used throughout Section 6. First, $DF(0) = \operatorname{antidiag}(p_1(0), q_0'(0), r_0(0))$, where $p_1$ is the $v$-coefficient of $P$, etc.; target scalings normalize $p_1(0) = q_0'(0) = r_0(0) = 1$, hence (at $m = 3$) $\kappa = \det DF(0) = -1$. Second, the source shears $z \mapsto z + y^m h(xy)$ act on the invariants as $v \mapsto v + w^m\, h(w)$ — the box-preserving part of the full *$v$-shear group* $v \mapsto v + h(w)$, $h \in \mathbb{C}[w]$ arbitrary, under which $\det M$ is exactly invariant (asserted). The full $v$-shear group is an analysis tool: one solves modulo shears first and imposes the polynomiality box last.

# 5 The weight system $(2, -1, -3)$: $A_1$-cone reduction and degree-1 classification

All claims in this section are proved by assertion in `scripts/search_213.py` and its module `jcqft/reduction_213.py`; the repository document `docs/SEARCH_213.md` gives the full account.

## 5.1 Why this weight system

For $\lambda.(x, y, z) = (\lambda^2 x, \lambda^{-1} y, \lambda^{-3} z)$ the orbit space has *two* orbifold strata instead of the single one of the Alpöge–Mathew system: the target $x$-axis has stabilizer $\mathbb{Z}_2$ (the same 2:1 mechanism that powers the counterexample) and the target $z$-axis has stabilizer $\mathbb{Z}_3$ (a free orbit mapping into it would be 3:1 — cube-root escape and a $\mathbb{Z}_3$ monodromy class, a candidate for a different global-anomaly class). Unlike the $(1, -1, -m)$ family the invariant ring is not free, so the reduction of Proposition 4.1 does not apply and was rebuilt.

## 5.2 Invariant theory and normal form (exact)

Monomials $x^i y^j z^k$ are invariant iff $2i - j - 3k = 0$; the invariant ring is the $A_1$ quadric cone

$$R \;=\; \mathbb{C}[u_1, u_2, u_3]/(u_2^2 - u_1 u_3), \qquad u_1 = xy^2, \quad u_2 = x^2 yz, \quad u_3 = x^3 z^2,$$

verified through a unique normal form for invariant monomials (exhaustively for degree $\leq 12$). The spaces of weight-$d$ polynomials are $R$-modules with verified syzygy-free splittings, e.g. $M_{-1} = yR \oplus xz\,\mathbb{C}[u_3]$ and $M_{-3} = zR \oplus y^3 \mathbb{C}[u_1]$. Invertibility of $DF(0)$ forces the component weights to be a permutation of $(2, -1, -3)$, and every candidate has the normal form

$$F \;=\; \big(\, x\,A(u), \;\; y\,B(u) + xz\,C(u_3), \;\; z\,D(u) + y^3 E(u_1)\,\big), \qquad DF(0) = \operatorname{diag}\big(A(0), B(0), D(0)\big),$$

with no polynomiality side conditions (every polynomial choice of $(A, B, C, D, E)$ assembles to a polynomial map).

### 5.3 Chart-based reduced Keller identity (exact)

The $\mathbb{C}^*$-action is free on $\{y \neq 0\}$, and on $\{x \neq 0,\ y \neq 0\}$ the slice parametrization

$$\Phi(t,p,q) \;=\; \Big(\frac{p}{t^2},\ t,\ \frac{q\,t^3}{p^2}\Big), \qquad t = y,\ p = u_1,\ q = u_2,$$

pulls the invariants back to $(p, q, q^2/p)$ — the cone in its chart $u_3 = u_2^2/u_1$. Writing $\tilde a = A(p,q,q^2/p)$, $\tilde b = B(\cdot) + \frac{q}{p}C(q^2/p)$, $\tilde e = \frac{q}{p^2}D(\cdot) + E(p)$, one gets $F \circ \Phi = (p\tilde a/t^2,\ t\tilde b,\ t^3\tilde e)$ and, by the chain rule,

$$\det DF \;=\; \Delta(p,q) \;=\; p^2\Big(2\,p\tilde a\,J_2(\tilde b,\tilde e) + \tilde b\,J_2(p\tilde a,\tilde e) - 3\,\tilde e\,J_2(p\tilde a,\tilde b)\Big), \tag{3}$$

with $J_2$ the Jacobian in $(p,q)$: a rational function of the two invariant chart coordinates alone, with denominator a pure power of $p$. The identity is proved with undetermined functions and re-verified against the raw 3D determinant and against an intrinsic normal-form determinant on the cone. Since $\Phi$ parametrizes a dense open set, $\det DF \equiv \kappa$ iff $\operatorname{numer}(\Delta - \kappa) \equiv 0$ in $\mathbb{C}[p,q]$ — a finite, *trilinear* polynomial system (degree $\le 1$ in each of the blocks $A \mid B, C \mid D, E$). Gauge: target scalings set $A(0) = B(0) = D(0) = 1$, $\kappa = 1$; the constant $xz$-shear sets $C(0) = 0$ (box-preservingly); $E(0)$ cannot be gauged away and is kept as an unknown.

The two stabilizer-jump mechanisms become pointwise conditions: a free orbit over $(p_0, q_0)$, $p_0 \neq 0$, maps 2:1 onto the target $x$-axis iff $\tilde b = \tilde e = 0 \neq \tilde a$ there, and 3:1 onto the target $z$-axis iff $\tilde a = \tilde b = 0 \neq \tilde e$; analogous conditions cover the $y = 0$ stratum, and the $x = 0$ stratum supports no mechanism. Any Keller solution with such a witness would automatically be a counterexample.

### 5.4 The degree-1 box: complete classification

**Theorem 5.1** (Degree-1 classification; exact, box-complete)**.** *In the box $A, B, D$ of degree $\le 1$ in $(u_1, u_2, u_3)$, $C = c_3u_3$, $E = e_0 + e_1u_1$ (canonical, syzygy-free), with the gauge above — 12 unknowns, 20 trilinear equations — the Keller variety is 2-dimensional and decomposes exactly (not just up to closure) as $V(I) = V_1 \cup V_2$:*

1. *$V_1$ (elementary $z$-shears): $F = (x,\ y,\ z + y^3(e_0 + e_1u_1))$, i.e. $z \mapsto z + y^3h(xy^2)$;*
2. *$V_2$ ($w$-shears): with $w = xz - \nu y$ and parameters $(\lambda, \nu)$,*

$$F = \big(x,\ y + \lambda x w^3,\ z + \lambda\nu w^3\big), \qquad w \circ F = w,$$

*an elementary shear in the coordinates $(x, w, z)$.*

*Both families are tame automorphisms with explicit polynomial inverses (verified by composition) and generic fiber cardinality* 1*. Moreover all six orbifold-mechanism systems (*2:1 *and* 3:1*, at the torus-normalized witness points and on the $y = 0$ stratum) are provably empty (Gröbner basis* = [1]*).*

The completeness proof is exact: six relations have powers inside the ideal $I$ (Gröbner reduction certificates), so they vanish on the whole variety, and the ideal element $c_3e_1^3 \in I$ splits the variety into $\{c_3 \neq 0\}$ (pinned to $V_2$), $\{e_1 \neq 0\}$ (pinned to $V_1$) and $\{c_3 = e_1 = 0\}$ (pinned to $V_1$ again). The residual 2-torus acts freely on witness points and preserves box and gauge, so the pointwise mechanism checks exhaust all witnesses.

In the enlarged *CE2 box* ($C \in \mathrm{span}\{u_3, u_3^2\}$, $E \in \mathrm{span}\{1, u_1, u_1^2\}$; 14 unknowns, 34 equations) the six mechanism emptiness queries are again all empty (exact, over $\mathbb{Q}$; longest single Gröbner basis $\approx 5$ min). A full variety decomposition of CE2 was not attempted, so in that box only the stabilizer-jump mechanisms are excluded — a Keller map could in principle identify two distinct free orbits; ruling that out needs the full decomposition, done only for the degree-1 box. (For Alpöge–Mathew the realized mechanism *is* the stabilizer jump, which motivates prioritizing it.)

### 5.5 The degree-2 box: an honest wall

The full degree-2 box (29 unknowns, 57 Keller equations plus witness conditions) is **unresolved, not ruled out**, after a serious tooling campaign whose obstruction is now precisely mapped (`scripts/deg2_213_elim.py`):

1. sympy's Buchberger: no query finishes ($> 25$ min each, also mod $p$).
2. msolve F4 [19] (under a 16 GB address-space cap): all six queries exceed the cap over $\mathbb{Q}$ — and, decisively, also mod three independent $\sim$30-bit primes, even after the exact reductions below; the memory blow-up is intrinsic to F4 on this system, not a coefficient-growth artifact.
3. Exact structural reductions (all asserted): the 57 Keller equations are triangular in the $A$-block with integer pivots in $\{2, \dots, 7\}$, so the $A$-block eliminates by a global polynomial substitution valid over any field of characteristic 0 or $p > 7$ (49 equations, 21 unknowns); the Rabinowitsch saturation variable is unnecessary because $\Delta = \tilde{a}X + \tilde{b}Y + \tilde{e}Z$ identically; witness equations eliminate 1–2 more unknowns. Net: 19–20 unknowns per query — still past the 16 GB cap for F4, even mod $p$.
4. Singular's truncated `std()` ladder [21] (degree bound; one-sided: finding 1 below the bound is an exact certificate): mod $p$, degree bounds 4 and 5 complete in under a minute with no certificate; bounds 6 and 7 exceed 20-minute budgets. Hence the unit certificate, if the ideal is the unit ideal, has degree $\geq 6$.

Plausible routes forward: much larger memory, further stratification, or numerical algebraic geometry (homotopy continuation) for discovery with exact certification of any find; see Section 12.

### 5.6 Verdict

The weight system $(2, -1, -3)$ *does* support nonlinear Keller maps — two 2-parameter families — but every Keller map in the degree-1 box is a tame automorphism, and both orbifold mechanisms are provably empty in the degree-1 and CE2 boxes. The $\mathbb{Z}_3$ stratum made this the most promising place to look for a new global-anomaly class; the answer in the searched boxes is a clean no-go. The contrast with $(1, -1, -2)$, where the Alpöge–Mathew defect exists already at low degree, suggests the defect needs the free invariant ring of the $(1, -1, -m)$ family, or larger boxes; the $A_1$ cone singularity appears to rigidify the low-degree Keller moduli. (Interpretation, clearly flagged as such.)

# 6 The weight system $(1, -1, -3)$: complete classification of the $v$-linear class

This is the $m = 3$ case of the uniqueness theorem (Section 7). Every identity and every emptiness claim is proved by assertion in `scripts/search_113.py` (default run ∼20 min; `-full` ∼1 h adds larger Gröbner boxes); the reduction machinery is Proposition 4.1 at $m = 3$.

## 6.1 Setup: why $(1, -1, -3)$, and the shape of the problem

For $\lambda.(x, y, z) = (\lambda x, \lambda^{-1}y, \lambda^{-3}z)$ the invariant ring is free — $\mathbb{C}[w, v]$ with $w = xy$, $v = x^3z$ — exactly as in the Alpöge–Mathew case $m = 2$. The target coordinates carry weights $(-3, -1, 1)$, so the target $a$-axis has stabilizer $\mathbb{Z}_3$: a free orbit mapping into it is 3:1 (cube-root escape, $\mathbb{Z}_3$ monodromy). Moreover *no* 2:1 *mechanism exists at all* in this weight system: every other coordinate axis and every coordinate pair has trivial stabilizer, so the only stabilizer-jump non-injectivity available is the 3:1 one. Since the Alpöge–Mathew counterexample is exactly a stabilizer-jump (2:1) map in the $m = 2$ member of the same family, $(1, -1, -3)$ is the natural place to ask whether the mechanism *transplants*, with the residual $\mathbb{Z}_2$ upgraded to $\mathbb{Z}_3$.

By Proposition 4.1, every candidate is $F = (P/x^3,\ Q/x,\ xR)$ with polynomiality box $j+3k \geq 3$ for the monomials $w^jv^k$ of $P$ and $j+3k \geq 1$ for those of $Q$; the gauge $p_1(0) = q_0'(0) = r_0(0) = 1$ fixes $\kappa = -1$.

The class searched is the *$v$-linear class*: $P, Q, R$ of degree $\leq 1$ in $v$ and *arbitrary* degree in $w$,

$$P = p_0(w) + p_1(w)\,v, \qquad Q = q_0(w) + q_1(w)\,v, \qquad R = r_0(w) + r_1(w)\,v,$$

with box/invertibility constraints $p_0 \in w^3\mathbb{C}[w]$, $q_0 \in w\,\mathbb{C}[w]$, $p_1(0) = q_0'(0) = r_0(0) = 1$. This is the exact analogue of the class containing the Alpöge–Mathew map at $m = 2$ (whose $P, Q, R$ are all $v$-linear). Since $\det M + 1$ has $v$-degree 2, the Keller condition is three ODE-type equations in $\mathbb{C}[w]$,

$$E_2 = 0, \qquad E_1 = 0, \qquad E_0 = \kappa = -1,$$

with (all asserted from $\det M$, not copied from $m = 2$)

$$E_2 = 2\big[\, q_1(p_1r_1)' - 2(p_1r_1)q_1' \,\big], \qquad E_0 = -3p_0(q_0'r_1 - q_1r_0') + q_0(p_0'r_1 - p_1r_0') + r_0(p_0'q_1 - p_1q_0'),$$

and $E_1$ the mixed analogue,

$$E_1 = -3p_0[Q, R]_1 - 3p_1[Q, R]_0 + q_0[P, R]_1 + q_1[P, R]_0 + r_0[P, Q]_1 + r_1[P, Q]_0,$$

where $[A, B]_i$ denotes the $v^i$-part of $J_2(A, B)$. Unlike the degree-boxed searches of Section 5, the $w$-direction here is *unbounded*: the classification below is a theorem for the whole class, not for a box — with one precisely-mapped exception (Section 6.4).

## 6.2 The Wronskian stratification

**Lemma 6.1** (Integration certificate; asserted)**.** $\big(p_1r_1/q_1^2\big)' = E_2/(2q_1^3)$. *Hence on* $\{E_2 = 0,\ q_1p_1r_1 \neq 0\}$ *the rational function* $p_1r_1/q_1^2$ *has vanishing derivative, so*

$$p_1r_1 \;=\; c\,q_1^2, \qquad c \in \mathbb{C}^\times,$$

*exactly; and* $q_1 = 0$ *or* $r_1 = 0$ *satisfy* $E_2$ *identically.*

Since $p_1(0) = 1$ forces $p_1 \neq 0$, the $v$-linear Keller problem stratifies as follows. In stratum D, the UFD solution of $p_1 r_1 = c q_1^2$ is $p_1 = a s^2 g$, $q_1 = s t g$, $r_1 = b t^2 g$ with $g = \gcd(p_1, r_1)$, $\gcd(s,t) = 1$, $ab = c$; $E_0$ is linear in $(p_1, q_1, r_1)$ with no derivatives of them, so $g \mid E_0 = \kappa$ and $g$ is a unit (absorbed).

| stratum | definition | verdict at $m = 3$ |
|---|---|---|
| A | $q_1 = r_1 = 0$ | tame family (all $w$-degrees) |
| B | $r_1 = 0,\ q_1 \neq 0$ | tame family (all $w$-degrees) |
| C | $q_1 = 0,\ r_1 \neq 0$ | empty (all $w$-degrees) |
| D0 | $s,t$ const ($p_1, q_1, r_1$ const) | empty (all degrees) |
| D1 | $t$ const, $s$ nonconst (*the AM-analogue*) | empty (all degrees) |
| D2 | $s$ const, $t$ nonconst | empty (all degrees) |
| D3 | $s,t$ nonconst, coprime, both squarefree | empty (all degrees) |
| $D_3'$ | $s$ or $t$ non-squarefree ($\deg p_1 \geq 4$ or $\deg r_1 \geq 4$) | empty in Gröbner boxes; open beyond |

*Cross-check at $m = 2$.* The same $E_2$-integration at $m = 2$ gives $p_1^2 r_1 = c\, q_1^3$, and the Alpöge–Mathew data satisfies it with $c = -1/27$: $p_1 = (1+w)^3 = s^3$, $q_1 = 3(1+w)^2 = 3s^2$, $r_1 = -1$ (asserted). That is, the Alpöge–Mathew map lives in the $m = 2$ analogue of stratum D1 — the stratum that is *empty* at $m = 3$. The machinery does not spuriously kill the known counterexample; the two weight systems genuinely differ.

## 6.3 The main theorem

**Theorem 6.2** (Complete classification of the $v$-linear class, $m = 3$; exact, all $w$-degrees, modulo the $D_3'$ gap of Section 6.4)**.** *In the gauge $p_1(0) = q_0'(0) = r_0(0) = 1$, $\kappa = -1$, the complete solution set of the Keller condition in the $v$-linear class is the one-function-plus-one-parameter family*

$$\boxed{P = p_0 + v, \qquad Q = w + b_0 P, \qquad R = 1, \qquad p_0 \in w^3\, \mathbb{C}[w], \quad b_0 \in \mathbb{C}.}$$

*Every member assembles to $F = \big(p_0(xy)/x^3 + z, \quad y + b_0 x^2 F_1, \quad x\big)$: an elementary $z$-shear followed by the target shear $b \mapsto b + b_0\, ac^2$ — a* tame automorphism*, with explicit polynomial inverse verified by composition and generic fiber cardinality* 1*. Strata C, D0, D1, D2 and D3 (with $s,t$ squarefree) are empty for every $w$-degree.*

*Proof sketch.* We go stratum by stratum. Each step combines an identity chain asserted in `scripts/search_113.py` with a one-line divisibility argument in $\mathbb{C}[w]$ (a nonzero constant has no nonconstant divisors).

*A.* $E_1 \equiv 0$ and $E_0 = -p_1(q_0 r_0)' = \kappa$, so $p_1 \mid \kappa$, giving $p_1 = 1$ (gauge); then $q_0 r_0 = w$ with $q_0(0) = 0$, $r_0(0) = 1$ forces $r_0 = 1$, $q_0 = w$; $p_0 \in w^3\mathbb{C}[w]$ stays free.

*B.* $E_1$ integrates $((p_1 r_0^2/q_1)' = r_0 E_1/q_1^2$, asserted) to $p_1 r_0^2 = c\, q_1$; then $E_0 = p_1\big[(p_0 r_0^3)/c - q_0 r_0\big]'$, forcing $p_1 = 1$, $r_0 \mid w$ with $r_0(0) = 1$ so $r_0 = 1$, and $q_0 = b_0 p_0 + w$, $q_1 = b_0 = 1/c$. Together with A (the case $b_0 = 0$) this gives the boxed family.

*C.* $(p_1/(q_0^4 r_1))' = E_1/(q_0^5 r_1^2)$ (asserted), so $E_1 = 0$ forces $p_1 = c\, q_0^4 r_1$ and hence $p_1(0) = 0$, contradicting $p_1(0) = 1$.

*D0* ($p_1, q_1, r_1$ nonzero constants $\alpha, \beta, \gamma$): $E_1$ integrates to a linear relation; substituting into $E_0$ gives an exact derivative $[-(2\alpha\gamma/\beta)V^2 - \alpha\delta V]' = \kappa$ with $V = q_0 - (\beta/\alpha)p_0$, $V(0) = 0$, $V'(0) \neq 0$. Then $V \mid \kappa w$ forces $V = \varepsilon w$, whose quadratic term forces $2\alpha\gamma/\beta = 0$: impossible.

*D1* ($p_1 = As^2$, $q_1 = Bs$, $r_1 = C$, $s$ nonconstant — the Alpöge–Mathew slot): the $v$-shear $h = -r_0/C$ sets $r_0 \equiv 0$; then $E_1$ is a linear ODE for $p_0$ with particular solution $p_0^* = (2A/B)sq_0$

and homogeneous solutions satisfying $Y^2 = e\, s^3$ (nonzero only if $s$ is a constant times a square). If $s$ is not: $E_0(p_0^*) = (2AC/B)\, q_0(s'q_0 - 2sq_0') = \kappa$ forces $q_0 = n$ constant and $s$ linear. If $s = cd^2$: $E_0$ factors with a factor $d \mid \kappa$, which kills it. The surviving sheared family $(s, q_0, p_0, r_0) = (s_0 + s_1 w,\ n,\ (2A/B)ns,\ 0)$ has $\kappa = (2AC/B)n^2 s_1 \neq 0$, so $n \neq 0$; but *un-shearing into the box* $(p_0(0) = q_0(0) = 0)$ forces $n = 0$. Empty.

*D2* ($p_1 = A$ constant): the shear $h = -p_0/A$ sets $p_0 \equiv 0$; $E_0 = -A(q_0 r_0)' = \kappa$ makes $q_0 r_0$ exactly linear, so one factor is constant. If $q_0 = \gamma$: $E_1$ forces $t$ linear and $r_0 = (2C\gamma/B)t$, but un-shearing needs $\mathrm{val}_w(p_0 = -Ah) \geq 3$, hence $h(0) = 0$, hence $q_0(0) = \gamma \neq 0$: contradiction with $q_0(0) = 0$. If $r_0 = \delta$: $E_1$ gives $t \mid \delta^2 t'$, so $t' = 0$: contradiction.

*D3* ($s, t$ nonconstant, coprime, squarefree): pass to the shear-invariant combinations $G_1 = tp_0 - asq_0$, $G_2 = btq_0 - sr_0$. Asserted identities express $E_1$ and a combination of $E_0, E_1$ through $G_1, G_2$ and $G_3 = btG_1 + asG_2$; $G_1 = 0$ or $G_2 = 0$ contradict $\kappa \neq 0$. Squarefree divisibility gives $s \mid G_1$, $t \mid G_2$; writing $Z = a\bar{g}_2 - b\bar{g}_1$ and the Wronskian $W = s't - st'$, one finds $E_1 = st(WZ - 2stZ')$ and $(Z^2 t/s)' \propto Z(2stZ' - WZ)$, forcing $Z^2 t = es$ and hence $Z = 0$. The $\kappa$-equation then reads $g(2stg' - Wg) = -b\kappa/(2a)$, so $g$ is a constant $c$ and $\kappa = (2a/b)c^2 W$: Keller solutions exist iff the Wronskian $W$ is a nonzero constant, and they form the single shear-orbit $p_0 = as(q_0 + c/b)/t$, $r_0 = t(bq_0 - c)/s$. But $tp_0 = as(q_0 + c/b)$ evaluated at $w = 0$ gives $p_0(0)t(0) = a\, s(0)\, c/b \neq 0$, while the box forces $p_0(0) = 0$: empty for every degree. □

*Remark* 6.3. The $v$-shear moves (arbitrary $h \in \mathbb{C}[w]$) are *not* box-preserving; the box is imposed at the end through the jet conditions at $w = 0$. This is exactly where the $m = 3$ obstruction lives (Section 6.6).

### 6.4 The gap: $D_3'$, and its closure in boxes

The two steps $s \mid s'G_1 \Rightarrow s \mid G_1$ and $t \mid t'G_2 \Rightarrow t \mid G_2$ in the D3 argument need $s, t$ squarefree. A non-squarefree $s$ or $t$ requires $\deg p_1 = 2 \deg s \geq 4$ or $\deg r_1 = 2 \deg t \geq 4$. This corner is closed by exact in-box Gröbner certificates (msolve [19], over $\mathbb{Q}$, under a 16 GB memory cap):

1. *Default run:* Keller + ($r_1$ has a nonzero coefficient) is the unit ideal in the box $\deg(p_0, p_1, q_0, q_1, r_0, r_1) \leq (4, 2, 3, 2, 2, 2)$ — covering strata C and D wholesale ($\sim$5 s per coefficient).
2. *`-full`:* the same in the medium box $\leq (5, 3, 4, 3, 3, 3)$, plus the four targeted non-squarefree parametrizations $s = (1 + \rho w)^2$ / $t = (n_0 + n_1 w)^2$ with the partner of degree $\leq 2$ and $\deg(p_0, q_0, r_0) \leq (6, 5, 4)$, each a direct emptiness query in 18–19 unknowns. Status (2026-07-24, 30 GB machine): the $s$-square/$t$-linear query is **empty** (exact, 649 s); the remaining three hit msolve's 16 GB F4 memory wall — the same wall as Section 5.5 — and remain formally unresolved in the large box. They *are* covered by the medium-box $r_1 \neq 0$ queries up to $\deg r_1 \leq 3$, so only the deg $= 4$ corner is genuinely open.

Beyond the boxes, $D_3'$ is *open*, though unpromising: the squarefree D3 analysis shows the entire stratum's solution set is one shear-orbit killed by the origin jet, and the non-squarefree variant only tightens the divisibility constraints. Independently of the gap, the classification is corroborated in-box by nilpotency certificates: on the $r_1 = 0$ slice of the small box, powers of the relevant coefficient combinations all lie in the Keller ideal, pinning the in-box variety exactly to the gauged family of Theorem 6.2 (with its box truncation).

### 6.5 The 3:1 mechanism is empty; no 2:1 exists

**Corollary 6.4** (Verified)**.** *A* 3:1 *stabilizer-jump witness needs a point* $(w_0, v_0)$ *with* $Q = R = 0 \neq P$ *on a free orbit* $\{x \neq 0\}$ *(the* $x = 0$ *plane supports no mechanism:* $F_3 \equiv 0$ *there and the induced 2D map is linear in its orbit parameter). By Theorem 6.2 every* $v$*-linear Keller map has* $R = 1$ *in the gauge —* $R$ *never vanishes, so no witness exists, for any* $w$*-degree. Independent pointwise Gröbner queries (the residual torus moves any witness to* $(1,1)$*,* $(1,0)$ *or* $(0,1)$*;* $(0,0)$ *is excluded by* $R(0,0) = 1$*) confirm emptiness in the small box. By weight arithmetic there is no* 2:1 *mechanism anywhere in this weight system, so the* $v$*-linear class of* $(1,-1,-3)$ *admits no orbifold-type counterexample at all.*

The family members survive the repository's non-properness prefilter only through its documented false-positive class of nonlinear automorphisms (module `jcqft/prefilter.py`), and injectivity is re-checked directly: explicit polynomial inverses by composition, and generic fiber cardinality 1 on samples.

**Headline: no counterexample.** Every Keller map of the $v$-linear class is a tame automorphism.

### 6.6 Why Alpöge–Mathew does not transplant: the exact numerology

At $m = 2$ the Alpöge–Mathew map sits in stratum D1 of its own family: $p_1 = s^3$, $q_1 = 3s^2$, $r_1 = -1$, $s = 1 + w$ — the same "$t$ constant" slot that at $m = 3$ produces the sheared family $q_0 = n$, $p_0 = (2A/B)ns$. The $m = 3$ box demands $\mathrm{val}_w\, p_0 \geq 3$ (from $j + 3k \geq 3$), while the family jets force $p_0(0) \neq 0$ unless $n = 0$, i.e. $\kappa = 0$. At $m = 2$ the box demands only $\mathrm{val}_w\, p_0 \geq 2$, *and* the corresponding equations carry a different derivative-weighting ($E_0^{m=2} = q_0 p_0' - 2p_0 q_0' + \dots$), which is exactly what lets $P_{\mathrm{AM}} = (1 + w)\big(v(1 + w)^2 + w^2(3w + 4)\big)$ thread the needle. The obstruction at $m = 3$ is *numerological and exact* — not a failure to search far enough.

### 6.7 Honest limitations

1. *$v$-degree.* The class searched is $v$-linear (the AM-analogue class). Ansätze of $v$-degree $\geq 2$ are untouched; for them the Keller condition has higher $v$-degree and the Wronskian stratification does not directly apply. (At $m = 2$ the known counterexample is $v$-linear; a heuristic, not an argument, for prioritizing the class.)

2. *The $D_3'$ gap* (Section 6.4): closed only in Gröbner boxes; open beyond, with the $\deg = 4$ corner the genuinely open part.

3. *Gauge bookkeeping.* The classification is stated in the gauge $p_1(0) = q_0'(0) = r_0(0) = 1$, $\kappa = -1$; un-gauged solutions are recovered by target/torus scalings, which change neither tameness, properness nor fiber counts.

## 7 Uniqueness for every $m \geq 3$

The $m = 3$ classification of Section 6 extends uniformly. Every identity below is asserted in `scripts/search_11m.py` (78 checks, $\sim$1 min default; `-full` $\sim$72 min), with $m$ a sympy `Symbol` wherever the statement is uniform and with $k$-symbolic parity branches ($m = 2k$ / $2k{+}1$) otherwise; spot-checks at $m = 2, 3, 4, 5, 7$. Full write-up: `docs/SEARCH_11M.md`.

**Theorem 7.1** (Uniqueness in the $v$-linear class; verified with gaps)**.** *For every integer $m \geq 3$, every Keller map in the $v$-linear class of $(1, -1, -m)$ is, up to gauge, the tame automorphism*

$$P = p_0 + v, \qquad Q = w + b_0 P, \qquad R = 1, \qquad p_0 \in w^m\, \mathbb{C}[w], \quad b_0 \in \mathbb{C}.$$

*Hence that class contains no counterexample and no $m$:1 orbifold covering for any $m \geq 3$, and the Alpöge–Mathew map ($m = 2$) is the unique member of its equivariant family within the $v$-linear class. The same $D_3'$ non-squarefree gap as at $m = 3$ remains, box-closed for $m \leq 5$.*

*$v$-quadratic extension at $m = 2$ (`scripts/search_vquad.py`; `docs/SEARCH_VQUAD.md`).* Raising the $v$-degree to 2 inside a finishable $\deg_w$ box ($\deg_w$ of the $v$-linear part $\leq (4, 3, 1)$, of each $v^2$ coefficient $\leq 1$) produces no new counterexample: any genuine $v^2$ term forces $R \equiv 1$, hence the map is tame by the Jung–van der Kulk theorem in the reduced plane; the $v$-linear slice is again tame $\cup\{\mathrm{AM}\}$. A cheap $\deg_v = 3$ probe shows the same pattern. Uniqueness of the defect is therefore stable under the first raise of $v$-degree at the AM weight.

*The general integrated constraint.* The $v^2$-coefficient of $\det M$ (with $m$ symbolic) is $E_2 = 2p_1' q_1 r_1 - (m{+}1) p_1 q_1' r_1 + (m{-}1) p_1 q_1 r_1'$, i.e. log-derivative exponents $(2, -(m{+}1), m{-}1)$. Integration gives

$$p_1^2\, r_1^{m-1} = c\, q_1^{m+1} \qquad (c \in \mathbb{C}^\times),$$

via the asserted certificate $(p_1^2 r_1^{m-1}/q_1^{m+1})' = p_1 r_1^{m-2} E_2 / q_1^{m+2}$. Anchors: $m = 2$ recovers $p_1^2 r_1 = c q_1^3$ (AM at $c = -1/27$); $m = 3$ recovers the square of Lemma 6.1.

*Stratum verdicts.* Strata A/B give the tame family and C, D0 are empty, all with $m$ symbolic. For D1–D3 the bookkeeping branches on parity of $m$; for $m \geq 4$ the $\kappa$-equations carry nonconstant killing factors — $u^{m-2}$ (D1 even), $u^{(m-3)/2}$ and $d^{m-2}$ (D1 odd), $s^{k-1}$ (D3 odd), $s^{m-2}$ (D3 even) — forcing a nonconstant polynomial to divide the constant $\kappa$. Thus for $m \geq 4$ the D-strata are empty *outright, with no box*. Only $m = 3$ needs the box-jet arguments of Section 6; it is the boundary case of the uniform mechanism.

*The exact $m = 2$ degeneracy.* Alpöge–Mathew sits in D1-even, where two degeneracies coincide at $m = 2$ and only there: (i) the killing factor $u^{m-2}$ is trivial; (ii) the even-only homogeneous $E_1$-direction $Y = \tilde{y}\, u^m$ exists, and AM needs $\tilde{y} = -1 \neq 0$. Asserted end-to-end: AM's sheared data ($p_0^{\mathrm{sh}} = (w{+}1)(w{+}2)$, $q_0^{\mathrm{sh}} = 6{+}4w$) plugs into the $\kappa$-formula and yields $E_0 = -2 = \kappa_{\mathrm{AM}}$ exactly. The machinery does not spuriously kill the known counterexample.

*Orbifold.* For every $m \geq 2$ (composite included) the only stabilizer-jump mechanism is $m$:1 on the weight $-m$ axis; the classification gives $R \equiv 1$, so it is empty. Pointwise Gröbner witnesses confirm in-box at $m = 4, 5$.

*Honest residue.* $D_3'$ beyond boxes (thresholds grow with $m$: $\deg \geq m{+}1$ odd / $2(m{+}1)$ even) remains open. One targeted $m = 5$ gap query hit the 16 GB F4 wall; its sibling closed EMPTY exactly, and all eight $m = 4/5$ medium-box queries closed EMPTY.

## 8 The 0D Witten index and the variational completion

The two remaining sections of results answer the two most natural QFT-side questions about the model: *what does supersymmetric localization see?* (Section 8.1), and *does the theory admit an action?* (Sections 8.2–8.4). Both answers sharpen the non-properness reading. All identities below are asserted in `scripts/witten_index.py` and `scripts/symmetric_search.py`; write-ups: `docs/WITTEN_INDEX.md`, `docs/SYMMETRIC_SEARCH.md`.

## 8.1 The Mathai–Quillen index jumps at the wall

Since $\det DF \equiv -2$, the restriction $F : \mathbb{R}^3 \to \mathbb{R}^3$ is an orientation-reversing local diffeomorphism, every $J$ is a regular value, and the signed solution count (the 0D Witten index [17]) is defined everywhere:

$$\deg(F, J) \;=\; \sum_{\phi \in F^{-1}(J)} \operatorname{sign}\det DF(\phi) \;=\; -N(J) \;=\; \begin{cases} -1, & p(J) > 0, \\ -3, & p(J) < 0. \end{cases} \tag{4}$$

The jump is itself an exact certificate of non-properness, independent of the escape-curve certificate of Theorem 3.2: a proper map extends to the one-point compactifications $S^3 \to S^3$ and its signed count at regular values is a single integer. Here the index changes chamber by chamber, which is possible only because vacua enter and leave through infinity in field space — index non-invariance *is* non-properness. Over $\mathbb{C}$, by contrast, the fiber count is 3 for every $J$ off $\{p = 0\}$: the wall is a real-locus phenomenon.

The index has a genuine SUSY realization. Although no superpotential exists (Section 8.2), the Mathai–Quillen completion [18] does, for arbitrary $F$: a nilpotent BRST charge and a $Q$-exact action whose Berezin integral over the fermionic pair $(\bar{\chi}, \psi)$ produces $\det DF$, verified generator-by-generator in an explicit Grassmann algebra. Its bosonic form

$$Z_\sigma(J) = (2\pi\sigma^2)^{-3/2} \int \det DF \; e^{-|F(\phi)-J|^2/2\sigma^2} \, d^3\phi$$

has the exact closed form $Z_\sigma(J) = -\,\mathbb{E}\big[N(J+\sigma\xi)\big]$ with $\xi \sim \mathcal{N}(0, \mathbf{1}_3)$: the Gaussian mollification of the index, finite for all $J$ and $\sigma > 0$, with $-3 < Z_\sigma < -1$ and $Z_\sigma(J) \to \deg(F, J)$ off the wall at the exact rate $e^{-\mathrm{dist}(J,\{p=0\})^2/2\sigma^2}$. On the wall the limit is $-2$; at the empty-fiber cusp orbit it is $-1$ — pure boundary-at-infinity contributions. (Direct quadrature matches the closed form to $\lesssim 5 \times 10^{-5}$; labelled numerical.) In SUSY language: the localization computation is exact and still $J$-dependent, because the compactness hypothesis under which the Witten index is deformation-invariant fails in exactly the way the counterexample is built to violate.

## 8.2 No action in three fields: the gradient no-go

Section 2 noted $DF \neq DF^{\mathsf{T}}$: as given, $F$ is not a gradient. This is not an artifact of the chosen frame (`scripts/symmetric_search.py`):

**Proposition 8.1** (No affine frame makes $F$ variational; verified)**.**

$$\big\{K \in \mathbb{C}^{3\times 3} \;:\; K\,DF(u) \textit{ symmetric for all } u\big\} = \{0\}.$$

*Consequently no invertible affine source/target change makes the Alpöge–Mathew map a gradient map. Moreover no affine normalization $x + H$ of $F$ has nilpotent $DH$ (the trace condition $\mathrm{tr}(C\,DF) = 3$ is an inconsistent linear system).*

The exclusion extends to the whole equivariant family: for every $m \geq 2$, the only gradient Keller maps equivariant for $(1, -1, -m)$ are linear maps and the explicit tame shear family $W = \alpha x^2/2 + \delta yz + \gamma y^{m+1}$ — outright for every $m \neq 3$ (weight branch table plus Wronskian/degree kills); at $m = 3$ the unique nontrivial branch $W = x^{-2} S(w, v)$ reduces to a rigid Monge–Ampère-type PDE, empty outright for $\deg_v S \leq 2$, with symbolic rigidity lemmas for the top coefficient rows and Gröbner-box closure at $\deg_v = 3, 4$. At $m = 2$ — the Alpöge–Mathew weights — the exclusion is complete: *the defect is non-variational in dimension 3.* Direct searches agree: the midpoint identity $\nabla W(a) - \nabla W(b) = \operatorname{Hess} W(\frac{a+b}{2})(a - b)$ kills $\deg W \leq 3$ in every dimension, and coefficient boxes are empty for $n = 2$ through $\deg W = 6$ and $n = 3$ through $\deg W = 4$ (full 25-coefficient box, exact over $\mathbb{Q}$). The honest wall is $n = 3$, $\deg W = 5$: 50 unknowns, beyond the 16 GB Gröbner cap even mod $p$.

### 8.3 Explicit variational counterexamples in dimension 6

De Bondt and van den Essen [16] reduced the Jacobian Conjecture to the symmetric case $F = \nabla W$ at the cost of doubling the dimension. Applying this to the Alpöge–Mathew map yields — to our knowledge for the first time — *explicit* variational Jacobian counterexamples, in dimension 6 (`scripts/symmetric_search.py`, all identities and witnesses asserted):

**Theorem 8.2** (Explicit symmetric counterexamples; verified)**.** (i) *The cotangent lift* $W_6(\varphi, \bar{\varphi}) = \bar{\varphi} \cdot F(\varphi)$ *— precisely the first-order action of Section 2 at* $J = 0$ *— satisfies* $\det \operatorname{Hess} W_6 \equiv -4$ *(*$\deg W_6 = 8$*, rational coefficients), and* $\nabla W_6 : \mathbb{C}^6 \to \mathbb{C}^6$ *is* 3:1 *over a rational point, with three explicit rational witnesses mapping to* $(\frac{5}{4}, 1, 1, -\frac{1}{4}, 0, 0)$*.* (ii) *The normalized twisted lift* $\widetilde{F} = \mathrm{id} + \nabla f_H$ *with* $f_H(x, y) = -i \sum_j H_j(x + iy)\, y_j$*,* $H = L^{-1}F - \mathrm{id}$*,* $L = DF(0)$*, satisfies* $\det(I + \operatorname{Hess} f_H) \equiv 1$ *and has three explicit witnesses over* $\mathbb{Q}(i)$ *with common image* $(0, 0, -\frac{1}{4}, 0, 0, 0)$*. In both cases non-injectivity transfers from* $F$ *by an exact block-determinant identity* $\det(I + \operatorname{Hess} f_H) = \det(I + DH)^2$*, re-proved from first principles in the script.*

Part (i) deserves emphasis: the action that Section 2 introduced as the standard remedy for $F$ not being a gradient is *itself* a symmetric counterexample — the auxiliary-field trick does not evade the defect, it inherits it, with the conjugate fields as cotangent directions. The minimal dimension of a symmetric counterexample is now pinned to $\{4, 5, 6\}$: dimension 3 is excluded in the controlled families and boxes above, and dimension 6 is achieved.

### 8.4 Why no *good* action exists: coercivity

The variational completion cannot be made thermodynamically sensible. A coercive nondegenerate real potential attains a minimum, forcing $\det \operatorname{Hess} W = \kappa > 0$; and $\det \operatorname{Hess} W = \text{const}$ forces the Hessian of the leading form of $W$ to be identically singular, so the leading form is parabolic, never strictly convex. The explicit $W_6$ fails coercivity in every available way at once: $\kappa = -4 < 0$, and $W_6$ is affine in $\bar{\varphi}$, hence unbounded below along every conjugate direction. So the sharpened statement is: *an action for the Alpöge–Mathew theory exists — one cotangent doubling away — but a coercive one does not*; the partition function $\int e^{-W}$ over the real slice can never be made absolutely convergent by choosing a better frame.

## 9 Classical-map invariants as Lagrangian data, and the first lattice lift

The results so far attach to one map a list of global invariants that no local or perturbative diagnostic sees. This section packages that list as *Lagrangian data* invariants of a theory on the same footing as anomaly coefficients and reports the first step off $D = 0$: an exact ultralocal/Galerkin probe and a certified homotopy-continuation computation showing that *wall-crossing survives* *kinetic mixing.* Scripts: `scripts/classical_map_invariants_probe.py` (35 assertions), `scripts/lattice_chamber.py` with the Julia driver `scripts/hc_lattice_chamber.jl` [20]; write-up `docs/CLASSICAL_MAP_INVARIANTS.md`.

### 9.1 The 0D dictionary

To a polynomial classical field map $F : \mathbb{A}^n_{\text{fields}} \to \mathbb{A}^n_{\text{sources}}$ we attach the package (AM values in brackets; tags per Section 1.3):

- I1. generic fiber degree [3; exact];
- I2. Galois group of the eliminant / geometric monodromy [$S_3$ exact for both; labelled basis permutations still numerical];
- I3. non-properness (Jelonek) divisor [$\{p = 0\}$; exact];
- I4. real chamber function $N(J)$ [3/1 by sign $p$; exact off the wall];
- I5. signed Brouwer/Witten index [$-N(J)$; exact; needs orientation];
- I6. observable-algebra defect: is $F^*$ an automorphism; module generators of the cokernel [$\{x, x^2\}$ missing; exact];
- I7. torus weight system, if equivariant [$(1, -1, -2)$; exact];
- I8. variationality flag: $\exists W$ with $F = \nabla W$, in some affine frame; coercivity of $W$ [no in $n = 3$, Proposition 8.1; exact].

I1–I3 and I6 are complex-algebraic; I4–I5 need the real structure (and I5 an orientation); I8 is the gate between Sections 8.2 and 8.3. For the Alpöge–Mathew map every slot is computed. The programmatic claim of `docs/AMPLITUDES_CONNECTION.md` §1.1  treat these as computable invariants of a Lagrangian  is thereby made concrete: the package is finite, machine-checkable, and (next subsections) stable under the smallest honest lifts toward $D \geq 1$.

### 9.2 Escape as a boundary divisor (weighted chart compactification)

Because the field weights $(1, -1, -2)$ are mixed-sign, ordinary weighted projective space is the wrong compactification. The minimal structure that turns the known escape curves into ordinary boundary points is the two-chart partial compactification $\overline{X} = U_0 \cup U_\infty$ of field space with $s = 1/x$, $\gamma = F_3$, and Cartier divisor $D_\infty = \{s = 0\}$ (`scripts/weighted_compactification.py`, 44 assertions; `docs/WEIGHTED_COMPACTIFICATION.md`). Exact: the AM escape curve extends regularly to $D_\infty$; the classical map extends to a polynomial morphism $\overline{F}\colon U_\infty \to \mathbb{A}^3$; and $\overline{F}(D_\infty) = \{p = 0\}$ set-theoretically — the Jelonek set is the image of the boundary. Ordinary $\mathbb{P}^3$ homogenization fails concretely (component degrees $(7, 6, 4)$; total-degree leading cone marks fake escapes). Interpretively, "vacua at infinity," Gribov-at-infinity, and second-type Landau singularities are ordinary boundary evaluations of $\overline{F}$ on $D_\infty$; no continuum $D \geq 1$ claim is made.

### 9.3 Ultralocal tensoring and the kinetic probe (exact)

Two exact stability statements (`scripts/classical_map_invariants_probe.py`):

*(i) The invariants tensor under sites.* For the ultralocal product $F^{\times N} : (\mathbb{R}^3)^N \to (\mathbb{R}^3)^N$ (independent copies; $N = 2, 3$ asserted): $\det D(F^{\times N}) = (-2)^N$, real fibers multiply ($N_{\mathrm{tot}} = \prod_i N(J_i)$), the non-properness set is the union of per-site walls, and the signed index is the product $\prod_i(-N(J_i))$. Adding sites does not lose the Lagrangian data.

*(ii) Linear kinetic mixing does not restore properness.* Let $F_\varepsilon(\phi_0, \phi_1) = \big(F(\phi_0) + \varepsilon(\phi_1 - \phi_0),\, F(\phi_1) + \varepsilon(\phi_0 - \phi_1)\big)$ two sites coupled by a discrete Laplacian, equivalently the two-mode Galerkin truncation of a 0+1 theory with ultralocal potential. The leading forms of $F_\varepsilon$ are unchanged (the mixing is linear), and on the equal-mode slice the kinetic term vanishes identically, so the AM escape curve a curve in the full $(\mathbb{R}^3)^2$ still goes to infinity with finite image: *$F_\varepsilon$ is non-proper for every $\varepsilon$* (exact). The product-fiber factorization, by contrast, is broken by mixing: a side effect, not a wash-out of the defect.

### 9.4 The lattice chamber function: wall-crossing survives kinetic mixing

The sharper question is whether the real chamber function $N_\varepsilon(J) = \#F_\varepsilon^{-1}(J) \cap (\mathbb{R}^3)^2$ remains non-constant at $\varepsilon > 0$ whether the walls survive as walls. This was computed by homotopy continuation [20] with three mutually cross-checking engines (a monodromy-stabilized master set over the joint parameter space, parameter homotopy from the nine exact $\varepsilon = 0$ product solutions, and fresh polyhedral solves), with reality and distinctness of every reported solution *certified* by interval arithmetic against exact rational input; completeness of the lists is numerical — and, as the exact computation of the next paragraph shows, incomplete at some points. Sample targets: $T_1$ (both sites in the $N$=3 chamber), $T_2$ (mixed), $T_3$ (both $N$=1). Certified real counts (complex counts in parentheses; dagger = HC lower bound, corrected exactly below):

| $\varepsilon$ | $T_1$ $(3 \times 3)$ | $T_2$ $(3 \times 1)$ | $T_3$ $(1 \times 1)$ |
|---|---|---|---|
| 0 (exact) | 9 (9) | 3 (9) | 1 (9) |
| 1/100 | 20 (52) | 12 (54) | 8 (66) |
| 1/10 | 16 (54) | 4 (54) | 10 (66) |
| 1/4 | $12^\dagger$ ($52^\dagger$) | 4 (54) | 6 (66) |
| 1/2 | 14 (46) | 8 (40) | 6 (60) |
| 1 | 12 (54) | 10 (54) | 8 (56) |
| 2 | 10 (50) | 8 (44) | 17 (61) |

*Verdict (probed range): the chamber function is non-constant at every probed $\varepsilon > 0$.* Structural findings:

1. *The ultralocal theory is the degenerate member of its own family.* The generic fiber degree over the joint parameter space is 66; at $\varepsilon = 0$ only 9 solutions are finite. Turning the kinetic coupling *on* brings dozens of solutions in from infinity, many real. Kinetic coupling does not suppress the boundary-of-field-space sector; it amplifies it.

2. *$F_\varepsilon$ leaves the Keller class* (exact): $\det DF_\varepsilon$ is non-constant for $\varepsilon > 0$ ($-2(1 - 2\varepsilon)(4\varepsilon^2 - 2)$ at the origin, field-dependent elsewhere). Lattice walls therefore mix fold-type (finite bifurcations) and escape-type (from infinity).

3. *Exact walls on the probed segment* (`scripts/lattice_discriminant.py`, 35 assertions; msolve eliminants over $\mathbb{Q}$). At $\varepsilon = 1/4$ on the line $J(t) = (1 - t)T_1 + t\,T_3$, the fold system

$\{F_\varepsilon(\phi) = J(t),\ \det DF_\varepsilon(\phi) = 0\}$ has exact eliminant $f(t) \in \mathbb{Z}[t]$ of degree 516 (squarefree, irreducible over $\mathbb{Q}$, 14 real roots in $(0, 1)$), and the escape locus is the degree-11 polynomial $e(t) = t(13t - 1)(3t + 1)q_y(t)q_x(t)$. Exact chamber counts interleaving the fold roots:

$$N:\ 18 \to 16 \to 14 \to 16 \to 14 \to 12 \to 10 \to 8 \to 6 \to 8 \to 6 \to 4 \to 2 \to 4 \to 6$$

— every jump $\pm 2$, every jump a fold. Escape is real but *pointwise*: $N(1/13) = 13$ between plateaus of 16, and the endpoint $T_1$ itself is an escape point ($N(0) = 14$ vs $N(0^+) = 18$; exact fiber degree $54 < 66$). The HC "odd jumps" $12 \to 13 \to 14$ of the previous paragraph — including the certified bracket $t \in (27/512, 7/128)$ — were completeness artifacts: neither $f$ nor $e$ has a root there, and the ideal degree inside the bracket is the generic 66. Structural contrast (exact): at $\varepsilon = 0$ all segment walls are escape-type; at $\varepsilon = 1/4$ all chamber walls are fold-type and escape only dips the count — kinetic mixing exchanges the wall mechanism.

Open, and stated as such: a uniform $\varepsilon_* > 0$ statement, more than two sites, the continuum limit, signed counts for the non-Keller $F_\varepsilon$, the fold hypersurface over a frozen source block (16 GB F4 wall), and $z$-completeness of $e(t)$.

### 9.5 The $D = 1$ Mathai–Quillen index: the jump survives the path measure

The sharpest form of the "vacua at infinity vs. measure suppression" question (open question C7 of the repository) asks whether a genuine $D = 1$ path measure kills the escaping equilibria that drive the 0D index jump of Section 8.1. In the finite-mode (Fourier) truncation the answer is now known (`scripts/d1_index_modes.py`, 37 assertions; `docs/D1_INDEX.md`): *the jump survives.*

Take periodic paths $q : [0, \beta] \to \mathbb{R}^3$ truncated to $M$ Fourier modes, the first-order flow $\dot{q} + F(q) - J$ as Nicolai map $G$ on the $3(2M{+}1)$-dimensional mode space, and $Z_\sigma = (2\pi\sigma^2)^{-n/2} \int \det DG\, e^{-|G|^2/2\sigma^2}$. Two exact statements (symbolic asserts):

- *$M = 0$ reduction:* the constant-path sector reproduces the 0D Mathai–Quillen integral exactly, with effective coupling $\sigma_{\mathrm{eff}} = \sigma/\sqrt{\beta}$ the kinetic normalization only rescales the coupling.
- *Saddle factorization:* at a constant-path zero $q^*$ (an equilibrium $F(q^*) = J$), $DG$ is block-diagonal over modes with $k$-block $\left(\begin{smallmatrix} A & \omega_k I \\ -\omega_k I & A \end{smallmatrix}\right)$, $A = DF(q^*)$, whose determinant is $\det(A^2 + \omega_k^2 I) = |\det(A + i\omega_k I)|^2 \geq 0$. The mode fluctuation determinant is therefore a $J$-independent *positive* spectator factor, $\operatorname{sign}\det DG(u^*) = \operatorname{sign}\det DF(q^*)$, and the $\sigma \to 0$ saddle sum is $\deg(F, J) = -N(J)$, *independent of $M$ and $\beta$* (spectral proviso $\pm i\omega_k \notin \operatorname{spec} DF(q^*)$ verified with gap 3.14 at every probed point).

Monte-Carlo integration (seeded, with error bars; proper-map controls give $-1.000 \pm 0.009$) confirms the exact prediction: at $\sigma = 0.025$ the chamber ratio $\bar{Z}(N{=}3)/\bar{Z}(N{=}1) = 2.998$–$3.006 \pm 0.009$ for all probed $(M, \beta) \in \{1, 2\} \times \{0.5, 1, 2\}$, and the near-wall crossover matches the 0D profile at $\sigma_{\mathrm{eff}}$ while the escaping pair sits at field strength $x^* = 8$. Honest gaps, stated as such: nonconstant periodic zeros of the flow are excluded *exactly* only for gradient force maps (for AM a 3840-start Newton probe finds none a probe, not a proof); the mass from infinity in mode space is probed ($\leq 3\%$, zero in the genuinely $D = 1$ channels at MC resolution), not bounded; and the $M \to \infty$/continuum order of limits is untouched. Within the truncation, the verdict stands: the path measure does not suppress the vacua at infinity  the kinetic term contributes only a positive spectator determinant. Together with Section 9.4, both halves of the smallest honest $D \geq 1$ lift (classical solution structure, and now the localized partition function) preserve the 0D defect.

### 9.6 Møller invertibility after kinetics (lattice and finite-mode $D = 1$)

In pAQFT the classical Møller map inverts the nonlinear field equation as a (typically formal) power series (Section 11.3). The zero-dimensional calibration is already sharp: that inverse may converge and still miss sheets at infinity. The first honest question off $D = 0$ is whether *kinetics* — lattice gradients or a $D = 1$ path measure — restore global invertibility of the classical map, or kill the escaping equilibria that drive the index jump of Section 8.1. Within the truncations computed above, the answer is negative on both counts.

On an $L$-site chain the ultralocal product $F^{\times L}$ tensors the classical-map invariants of Section 9.1 exactly. The kinetic deformation $F_\varepsilon$ of Section 9.4 keeps $N_\varepsilon$ non-constant for every probed $\varepsilon > 0$, so $F_\varepsilon$ stays non-injective after mixing; at $\varepsilon = 1/4$ on the probed segment the exact fold eliminant of degree 516 accounts for all chamber walls (escape only dips the count pointwise). A formal or convergent Møller series about the ultralocal vacuum therefore continues, after kinetics, to describe at most one local sheet of a still multi-sheeted classical map. Likewise the finite-mode Mathai–Quillen index of Section 9.5 remains $-N(J)$: path-space kinetics, in that truncation, do not single-value the classical inverse.

Upgrading a formal Møller map to a convergent one is still not the missing step toward a globally single-valued classical inverse: after the first kinetic deformations one still needs an independent properness (or degree) input. No statement is made about the continuum limits $M \to \infty$ or lattice spacing $a \to 0$, nor about interacting theories in $D \geq 2$.

## 10 The 0D Buchholz–Fredenhagen caricature: the dynamical relation sees the wall

Section 11.4 poses the exercise of formulating the 0D caricature of the Buchholz–Fredenhagen $S(f)$ relations [14] for this $F$. It is now carried out (`scripts/bf_caricature.py`, 38 assertions; `docs/BF_CARICATURE.md`), with a split verdict.

*The causal half trivializes, provably.* On a one-point spacetime causal disjointness of supports forces one of the two outer supports to be empty, and every allowed instance of the factorization relation $S(f+g+h) = S(f+g)S(g)^{-1}S(g+h)$ is a group-theoretic tautology; the only instance with content is exactly the one 0D causality excludes. This was the anticipated risk  it is now a verified statement rather than a suspicion, and it isolates what the 0D model can and cannot probe of [14].

*The dynamical half survives and is nontrivial.* Because no potential exists in three fields (I8, Proposition 8.1), the only Lagrangian implementing $F(\phi) = J$ is the first-order $L = \bar{\phi}\cdot(F(\phi)-J)$; the BF field-shift relation applied to the antifield $\bar{\phi}$ gives $\delta L(\beta) = \beta \cdot (F(\phi) - J)$ *exactly* ($L$ is affine in $\bar{\phi}$  no higher corrections), and multiplicative (pure-state) evaluations compatible with the relation are precisely the characters of the *fiber algebra*

$$A_J \; = \; \mathbb{C}[x, y, z] \,/\, (F - J).$$

The surviving BF-datum in 0D is the bundle of fiber algebras over source space together with its parallel transport, and the invariants of Section 9.1 occupy precise slots in it:

- the **wall** $\{p = 0\}$ is the rank-jump locus of the bundle ($\dim_{\mathbb{C}} A_J = 3 \to 1 \to 0$, the zero ring at the empty-fiber cusp orbit) *and* the pole divisor of every single-valued sheet-separating datum (the separator coefficients of Section 3 carry the factor $p$ verbatim);

- the $1 \leftrightarrow 3$ **count** is the number of characters of the real $C^*$-fiber $\mathbb{C}^{N(J)}$;

- $S_3$ is the holonomy of the parallel transport: wall meridians act by transpositions, the cusp loop by the order-3 Coxeter element (Exact group-level; labelled basis still numerical); and there is provably *no global deck action* the eliminant is irreducible with $S_3$ Galois group, so the degree-3 extension is non-normal and $\mathrm{Aut}(L/K) = 1$ (exact). "Deck transformation" formulations of the sheet symmetry are therefore wrong for this map; only transport holonomy survives.

The sharpest statement is an *obstruction dichotomy* (exact): any BF-style assignment $J \mapsto S(J)$ satisfying the 0D dynamical relation and resolving the classical sectors is multi-valued (the holonomy is transitive; no rational section of the cover exists) or singular on $\{p = 0\}$ (single-valued data factor through the trace subalgebra, whose sheet-separating elements have poles exactly on the wall). Conversely, everything single-valued and pole-free is a pullback through $F^*$ and is blind to the sectors. And the construction *collapses correctly*: for a proper Keller map (a tame automorphism control) it yields the trivial rank-1 bundle trivial holonomy, no wall so the caricature re-expresses non-properness as (rank jump, holonomy, pole divisor) of algebraic $S$-data rather than presupposing it.

The damped and Mathai–Quillen integrals of Sections 8.1–9.5 supply canonical *numerical* sections built from the character theory of $A_J$ (unsigned count $N(J)$, signed count $-N(J)$), finite for every source and blind to algebraic sector choice. In this sense the physical generating function $Z(J)$ is the single-valued shadow of the fiber-algebra bundle — an AQFT-flavoured packaging of the same Exact defect that the observable pullback $F^*$ detects algebraically.

### 10.1 BV antifield reading of the first-order action

The Alpöge–Mathew map admits no potential in three fields (Proposition 8.1): there is no $W$ with $\nabla W = F$. The only Lagrangian implementing $F(\phi) = J$ is therefore the first-order density $L = \bar{\phi} \cdot (F(\phi) - J)$, already used by Abdesselam [6] and forced here by the variationality flag I8. In Batalin–Vilkovisky language $\bar{\phi}$ is the antifield of $\phi$, and $L$ is the classical master-action density for $F - J = 0$. Two Exact faces of the same object appear elsewhere: the cotangent lift $W_6 = \bar{\varphi} \cdot F(\varphi)$ is the variational counterexample of Section 8.3 ($\det \mathrm{Hess}\, W_6 \equiv -4$, necessarily non-coercive), and antifield shifts of $L$ are exactly what force the fiber algebra $A_J$ in the dynamical relation above.

The constant Jacobian $\det DF \equiv -2$ makes the one-loop Faddeev–Popov / Berezin determinant field-independent, and the classical master equation for $L$ is the Koszul consistency of $F - J = 0$ — both are local. Neither forces the antifield equations to select a unique sheet: the fiber of $F$ is generically three points, the geometric monodromy is $S_3$, and $\{p = 0\}$ is the jump locus of $\dim A_J$. In slogans adapted to BV: *the master equation can hold with constant FP determinant while the antifield fiber bundle still carries nontrivial holonomy.* Global invertibility of the classical BV complex is an independent axiom, equivalent here to properness of $F$. The Mathai–Quillen BRST charge of Section 8.1 localizes to $-N(J)$ and *sees* the wall; it does not remove it. We claim nothing about continuum BV actions or BV anomalies in $D \geq 1$.

# 11 Implications for AQFT and pAQFT

This section is the interpretive core of the paper. The underlying facts are the exact statements of Sections 3–7; the contact with AQFT/pAQFT is *structural calibration*, not a theorem about $D \geq 1$ nets. We separate what can be said from what cannot, following the ledger of `docs/QFT_IMPLICATIONS.md`.

## 11.1 The defect and its name

In QFT language the Alpöge–Mathew phenomenon is a *non-properness defect*: a global consistency condition — properness of the classical field map, equivalently local constancy of the preimage count $N(J)$ — that is strictly stronger than every local criterion (the nonvanishing, indeed constant, Jacobian) and that standard field-redefinition arguments implicitly assume. It is not an anomaly in the strict cohomological sense (no symmetry is broken, nothing is loop-generated), but it shares the defining pattern of global anomalies: a manipulation valid by every local criterion fails through the global structure of configuration space. Its closest relative is the modern notion of anomalies in parameter space: dialing the external source around a closed cycle permutes the solution branches (the $S_3$ monodromy — Exact at group level, `scripts/certified_monodromy.py`), mediated entirely through infinity in field space. The exact observable-algebra statement of Section 3 is the algebraic face of the same defect: $F^*$ is a monomorphism but not an automorphism of the observable algebra; the two missing module generators $x, x^2$ are the vacuum separators; and their separator coefficients carry the escape polynomial $p$ verbatim.

A second, related lesson is about the *kind* of non-perturbative effect at work. The standard catalogue centres on finite-action saddle points (instantons). Here the extra sectors are not saddles at finite field values: they live on the boundary at infinity of field space and communicate with the finite region only across the non-properness hypersurface $\{p = 0\}$ — transitions "through infinity," realized exactly by the escape curve of Theorem 3.2. Whether an analogous boundary-of-field-space mechanism can survive in a genuine functional-integral setting is open; the example establishes that at the level of classical solution manifolds the mechanism exists and is compatible with completely trivial loop structure.

## 11.2 Haag–Kastler AQFT: algebras over field coordinates

Haag–Kastler AQFT [8] takes the *algebras of observables*, not field coordinates, as the primary data; fields are regarded as interchangeable "coordinates" on the theory (Borchers classes). The counterexample turns this philosophical preference into a theorem-sized fact in the only setting we control completely. The pullback

$$F^* : \ \mathbb{C}[a, b, c] \longrightarrow \mathbb{C}[x, y, z], \qquad F^*(\mathcal{O}) = \mathcal{O} \circ F,$$

is an injective algebra homomorphism that is *not surjective* (Theorems 3.1, 3.2): the polynomial observables of the "redefined field" $F(\varphi)$ form a proper subalgebra of the observables of $\varphi$, of index measured by the degree-3 field extension with Galois closure group $S_3$. So: a polynomial field redefinition with invertible propagator and constant unit Jacobian can be a monomorphism, but not an automorphism, of the observable algebra. Any framework whose objects are algebras and whose morphisms must be checked for surjectivity is structurally protected against the mistake this example punishes; any framework that treats "invertible field redefinition" as certified by the Jacobian is not. This is, we believe, the cleanest AQFT-flavoured lesson of the counterexample, and it is exact.

### 11.3 pAQFT (Fredenhagen–Rejzner): calibrating the deferral

In perturbative algebraic QFT [9, 10, 11] interacting observables are constructed from free ones by Møller-type maps built as formal power series in the coupling and $\hbar$; classical inversions of nonlinear field equations enter through the (classical) Møller map, and invertibility holds automatically at the formal-series level. Nothing in the counterexample exhibits an inconsistency in this: the constructions are internally coherent as formal deformation quantization, and their formal character is a *stated feature* of the framework. What the counterexample adds is a calibration of the worst case of that deferral, sharper than previously available: the formal inverse here is not merely consistent but *convergent*, analytic on a neighbourhood of the vacuum, and satisfies every identity perturbation theory can formulate — and still describes only one of three solution sectors, the others being invisible at every order because they sit at infinite field strength. So "upgrade formal to convergent" is *not* the missing step between pAQFT and a non-perturbative construction; the missing input is global (properness of the classical dynamics), and it must come from outside perturbation theory. Relatedly, statements of *perturbative agreement* (independence of the split of the action into free and interacting parts [12, 13]) are field-redefinition-adjacent moves established as formal-series identities; the example is a reminder of exactly which global questions such identities do not decide.

The same point bears on field redefinitions in renormalization theory more broadly. Arguments that treat a polynomial redefinition with $\det DF = \text{const}$ as an exact change of variables in a functional integral or a lattice measure — "$\det DF = 1$, so the measure is preserved" — use a false step: the pushforward under a non-injective $F$ overcounts sheets. Global injectivity is an honest hypothesis that must be verified, not read off from the Jacobian. Purely perturbative equivalence statements are untouched.

### 11.4 Non-perturbative algebraic constructions

The Buchholz–Fredenhagen $C^*$-algebraic approach [14] builds interacting dynamics from unitaries $S(f)$ subject to causal factorization relations, bypassing formal series altogether. Nothing in a 0D polynomial map bears on the correctness of that program. The counterexample instead supplies the minimal instance of a phenomenon any non-perturbative framework must *represent* somewhere in its data: classical dynamics whose solution count jumps (here $1 \leftrightarrow 3$) as an external source crosses an algebraic wall, with the extra solutions entering from infinity in field space and permuted by an $S_3$ monodromy under cycles of the source. The well-posed exercise of formulating the 0D caricature of the $S(f)$ relations for this $F$ is carried out in Section 10: the causal half trivializes (provably), while the dynamical half forces the bundle of fiber algebras in which the wall, the sheet count, and the $S_3$ holonomy all acquire exact algebraic addresses.

### 11.5 Rigidity of the defect: what the sibling searches say

The $(1, -1, -3)$ grading was the closest sibling of the Alpöge–Mathew theory with a $\mathbb{Z}_3$ orbifold stratum in place of $\mathbb{Z}_2$ — the candidate for a cube-root-escape global anomaly class. The verdict (Sections 6, 7) is a clean no-go in the AM-analogue class: the unit-determinant constraint is compatible with this grading only through shear-type field redefinitions; the theory admits no non-properness defect there and no $\mathbb{Z}_3$ vacuum-escape monodromy. Combined with the

$(2,-1,-3)$ no-go of Section 5 and the uniqueness theorem of Section 7, the pattern is that the Alpöge–Mathew defect is *rigidly attached to the* $m = 2$ *numerology*: neither enlarging the orbifold group, deforming the invariant geometry, nor moving to any other $m \geq 3$ reproduces it. For $m \geq 4$ the failure is an exact arithmetic incompatibility that needs no search box at all; $m = 3$ is the boundary case. Within the $v$-linear class the defect is a delicate coincidence of the $(1,-1,-2)$ weight arithmetic — the killing factor $u^{m-2}$ trivializing together with the even-only homogeneous $E_1$-direction.

For the QFT reading this is essential: it means the non-properness defect is *not* a generic feature of every constant-Jacobian theory, but a rare, exactly characterisable pathology. Frameworks that check global invertibility of field redefinitions are not being asked to police an open dense set of pathologies; they are being asked to exclude a delicate coincidence that, in the equivariant family we can control, sits only at $m = 2$.

### 11.6 What the model does not imply

Plainly, and to resist overstatement:

- *Nothing about UV renormalization.* The model is zero-dimensional; there are no short-distance singularities, no counterterms, no running. The phenomenon is invariant under everything renormalization theory cares about.
- *Nothing about Borel summability* of $\phi^4_2$ or $\phi^4_3$. Those are loop/large-order phenomena; our series is convergent. The two situations are disjoint.
- *Nothing about the core* $D = 4$ *difficulties.* Suspected triviality of $\phi^4_4$ and the Yang–Mills existence/mass-gap problem are ultraviolet and measure-theoretic. The counterexample does not bear on them.
- *No spacetime, no net, no causality, no Hilbert space.* Analogies between "extra solution sheets" and "sectors" are structural motivation, not theorems about $D \geq 1$ AQFT.

The honest headline is narrower and still interesting: *constant-Jacobian polynomial dynamics can hide non-perturbative sectors at infinity, and no amount of perturbative or even locally-analytic information detects them — but in the equivariant family* $(1,-1,-m)$ *this pathology is unique to* $m = 2$.

## 12 Outlook

Next probes, ordered by expected leverage for the QFT reading:

1. *The continuum* $D = 1$ *index.* Section 9.5 settles the finite-mode truncation (the jump survives, with the mode determinant an exact positive spectator); the remaining question is the $M \to \infty$/continuum order of limits, and an exact exclusion of nonconstant periodic flow zeros for non-gradient force maps.
2. *Exact lattice walls.* Section 9.4 computes the exact fold and escape polynomials of $F_\varepsilon$ on the probed segment at $\varepsilon = 1/4$ (degree-516 irreducible fold eliminant; all chamber walls fold-type; escape pointwise). Still open: the fold hypersurface over a frozen source block, a wall-crossing sign formula, other $\varepsilon$/segments, and $z$-completeness of the escape factor list.

3. *Higher $v$-degree / other $m$.* The $v$-quadratic class at $m = 2$ is settled in-box (only tame + AM; Section 7). Still open: full $\deg_v \geq 3$ boxes, $v$-quadratic at $m \geq 3$, and four-field gradings.

4. *Closing the $D_3'$ gap.* A structural argument removing the squarefree hypothesis would make Theorem 7.1 fully gap-free, strengthening the uniqueness claim that underwrites the "rare pathology" reading.

5. *Four-field gradings.* Richer stabilizer patterns and orbifold strata; the natural place to look for a genuinely different global-anomaly class once the three-field family is exhausted.

6. *The degree-2* $(2, -1, -3)$ *box.* Still beyond direct Gröbner methods on 30 GB hardware (Section 5.5); further stratification is needed before homotopy continuation is practical.

7. *The minimal symmetric counterexample.* Theorem 8.2 and the dimension-3 no-go of Section 8.2 pin the minimal dimension of a variational Jacobian counterexample to $\{4, 5, 6\}$; the $n = 3$, $\deg W = 5$ box and the $n = 4, 5$ question are the concrete open ends, along with a real (rather than $\mathbb{Q}(i)$) form of the twisted lift.

## 13 Reproducibility

All scripts run from the repository root as `.venv/bin/python scripts/<name>.py`; every displayed identity is an `assert` in the cited script. Runtimes are indicative (single desktop machine, 30 GB RAM).

| script | claim(s) verified | runtime |
|---|---|---|
| `verify_counterexample.py` | det $DF \equiv -2$; triple fiber (1) | seconds |
| `branch_locus.py` | eliminant cubic (2); wall $\{p = 0\}$; exact fibers | seconds |
| `monodromy.py` | $S_3$ sheet monodromy (labelled; numerical) | minutes |
| `certified_monodromy.py` | Mon $= S_3 = B_3 \twoheadrightarrow S_3$ (Exact, B6) | $\sim$1 s |
| `weighted_compactification.py` | escape chart: $\overline{F}(D_\infty) = \{p = 0\}$ (Exact) | $\sim$2 s |
| `missing_observables.py` | Theorems 3.1, 3.2 | $\sim$15 s |
| `reduction_113.py` | Proposition 4.1 ($m \leq 4$; boxes at $m = 3$) | $\sim$4 s |
| `search_213.py` | Section 5: identity (3); Theorem 5.1 | $\sim$40 s |
| `search_213.py -full` | CE2-box mechanism emptiness | $\sim$8 min |
| `deg2_213_elim.py` | degree-2 box reductions and capped screens (Section 5.5) | minutes |
| `search_113.py` | Section 6: Lemma 6.1, Theorem 6.2, Corollary 6.4 | $\sim$20 min |
| `search_113.py -full` | medium-box and targeted $D_3'$ certificates (Section 6.4) | $\sim$1 h |
| `search_11m.py` | Section 7: Theorem 7.1 (78 assertions) | $\sim$1 min |
| `search_11m.py -full` | medium-box / targeted $D_3'$ certificates for $m \leq 5$ | $\sim$72 min |
| `search_vquad.py` | Section 7: $v$-quadratic box at $m = 2$ (only tame + AM) | $\sim$6 min |
| `witten_index.py` | Section 8.1: index (4), jump certificate, MQ completion, closed form of $Z_\sigma$ | $\sim$10 s |
| `symmetric_search.py` | Section 8: Proposition 8.1, Theorem 8.2, gradient no-go, boxes, coercivity (72 assertions) | $\sim$3.5 min |
| `classical_map_invariants_probe.py` | Section 9.3: ultralocal tensoring; $F_\varepsilon$ non-proper for every $\varepsilon$ (35 assertions) | $\sim$2 s |
| `lattice_chamber.py` | Section 9.4: certified lattice chamber counts (needs Julia + [20]) | $\sim$3.5 min |
| `lattice_discriminant.py` | Section 9.4: exact fold/escape polys on the probed segment (35 assertions; needs msolve) | $\sim$6 min |
| `d1_index_modes.py` | Section 9.5: $M$=0 reduction, saddle factorization, MC jump ratio (37 assertions) | $\sim$4.5 min |
| `bf_caricature.py` | Section 10: vacuity of causal factorization, fiber-algebra bundle, obstruction dichotomy (38 assertions) | $\sim$4 s |


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## References


[1] L. Alpöge, announcement of a counterexample to the Jacobian Conjecture in dimension 3 (with the question posed by A. Mathew; the map produced by a large language model), pub-

lic announcement on X (`@__alpoge__`), July 19, 2026. A Lean 4 formalization is available at `github.com/deancureton/jacobian`.

[2] O.-H. Keller, *Ganze Cremona-Transformationen*, Monatsh. Math. Phys. **47** (1939) 299–306.

[3] H. Bass, E. H. Connell, D. Wright, *The Jacobian conjecture: reduction of degree and formal expansion of the inverse*, Bull. Amer. Math. Soc. (N.S.) **7** (1982) 287–330.

[4] D. Wright, *Formal inverse expansion and the Jacobian conjecture*, J. Pure Appl. Algebra **48** (1987) 199–219.

[5] A. van den Essen, *Polynomial Automorphisms and the Jacobian Conjecture*, Progress in Mathematics **190**, Birkhäuser, Basel (2000).

[6] A. Abdesselam, *The Jacobian conjecture as a problem of perturbative quantum field theory*, Ann. Henri Poincaré **4** (2003) 199–215. arXiv:math/0208173.

[7] A. Abdesselam, V. Rivasseau, *Trees, forests and jungles: a botanical garden for cluster expansions*, in *Constructive Physics*, Lecture Notes in Physics **446**, Springer (1995) 7–36. arXiv:hep-th/9409094.

[8] R. Haag, D. Kastler, *An algebraic approach to quantum field theory*, J. Math. Phys. **5** (1964) 848–861.

[9] K. Fredenhagen, K. Rejzner, *Perturbative algebraic quantum field theory*, in *Mathematical Aspects of Quantum Field Theories*, Springer (2015) 17–55. arXiv:1208.1428.

[10] K. Fredenhagen, K. Rejzner, *Quantum field theory on curved spacetimes: Axiomatic framework and examples*, J. Math. Phys. **57** (2016) 031101. arXiv:1412.5125.

[11] K. Rejzner, *Perturbative Algebraic Quantum Field Theory*, Mathematical Physics Studies, Springer (2016).

[12] S. Hollands, R. M. Wald, *Conservation of the stress tensor in perturbative interacting quantum field theory in curved spacetimes*, Rev. Math. Phys. **17** (2005) 227–311. arXiv:gr-qc/0404074.

[13] N. Drago, T.-P. Hack, N. Pinamonti, *The generalised principle of perturbative agreement and the thermal mass*, Ann. Henri Poincaré **18** (2017) 807–868. arXiv:1502.02705.

[14] D. Buchholz, K. Fredenhagen, *A $C^*$-algebraic approach to interacting quantum field theories*, Commun. Math. Phys. **377** (2020) 947–969. arXiv:1902.06062.

[15] Z. Jelonek, *The set of points at which a polynomial map is not proper*, Ann. Polon. Math. **58** (1993) 259–266.

[16] M. de Bondt, A. van den Essen, *A reduction of the Jacobian conjecture to the symmetric case*, Proc. Amer. Math. Soc. **133** (2005) 2201–2205.

[17] E. Witten, *Constraints on supersymmetry breaking*, Nucl. Phys. B **202** (1982) 253–316.

[18] V. Mathai, D. Quillen, *Superconnections, Thom classes, and equivariant differential forms*, Topology **25** (1986) 85–110.

[19] J. Berthomieu, C. Eder, M. Safey El Din, *msolve: A library for solving polynomial systems*, Proceedings of ISSAC 2021, ACM (2021) 51–58.

[20] P. Breiding, S. Timme, *HomotopyContinuation.jl: A package for homotopy continuation in Julia*, Mathematical Software – ICMS 2018, Springer LNCS **10931** (2018) 458–465.

[21] W. Decker, G.-M. Greuel, G. Pfister, H. Schönemann, SINGULAR — A computer algebra system for polynomial computations, https://www.singular.uni-kl.de.

[22] A. Meurer et al., *SymPy: symbolic computing in Python*, PeerJ Computer Science **3** (2017) e103.